

Airborne Detection of Crevasses Using Ground Penetrating Radar Technology

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List of Abbreviations

Abbreviation	Full-Form
ABS	Acrylonitrile Butadiene Styrene
ADC	Analog to Digital Converter
CCW	Counter-Clockwise
CFAR	Constant False-Alarm Rate
COCO	Common objects in Context (Dataset)
CPU	Central Processing Unit
CSV	Comma Separated Values
CW	Clockwise
DFL	Distribution Focal Loss
EDIA	Equality, Diversity, Inclusion and Accessibility
FFT	Fast Fourier Transform
FPS	Frame per Second
FMCW	Frequency Modulated Continuous Wave
GPR	Ground Penetrating Radar
GPU	Graphics Processing Unit
GUI	Graphic User Interface
LED	Light Emitting Diode
MAP	Mean Average Precision
ML	Machine Learning
PLA	Polylactic Acid
PVC	Polyvinyl chloride
RAM	Radar Absorbing Material
R-CNN	Region-based Convolutional Neural Network
RF	Radio Frequency
RUT	Range-cell Under Test
SAR	Synthetic Aperture Radar
SCPI	Standard Commands for Programmable Instruments
SDR	Software Defined Radio
SSD	Single-Shot Detector
SSPOC	Small Scale Proof of Concept
UAV	Unmanned Aerial Vehicle
URDF	Unified Robot Description Format
VCO	Voltage Controlled Oscillator
VNA	Vector Network Analyser
YOLO	You Only Look Once (Object Detector)

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Executive Summary

The project is motivated by the inherent risks and logistical problems involved in detecting the locations of subsurface crevasses within polar regions. These crevasses can be hazardous as they are often hidden by a layer of snow or ice, which can collapse under the pressure of a person or vehicle. This causes severe hazards for researchers, explorers, and rescue operations, necessitating reliable detection methods. During an initial stakeholder meeting Eef van Dongen, a polar researcher, was asked about the severity and scale of the problem and revealed “Yes, it happens. I knew a famous scientist who fell into a crevasse and died. You always need to be careful, even in areas where there are usually no crevasses”[1]. Traditional crevasse detection approaches such as manual probing, visual inspection, and helicopter-mounted radar survey are often dangerous, expensive, or logistically demanding. Consequently, there is a need for an efficient, inexpensive, and safe alternative capable of accurately detecting and visualising crevasses beneath the ice surface.

The aim of the project is to confirm the validity of airborne detection of crevasses and attempt to build a specialised system. This includes designing, implementing, and validating a lightweight, unmanned aerial vehicle (UAV) mountable Ground Penetrating Radar (GPR) prototype capable of providing real-time detection and visualisation of subsurface crevasses. However, for a complete proof of concept, this also requires demonstrating integrated drone control and automated detection in an intuitive form-factor.

The objectives comprise of several interrelated sub-tasks necessary to achieve this overarching aim. The primary objective addresses the GPR hardware which is made up of two parts. The first being a small-scale proof of concept utilising an existing low power radar device, the Anritsu MS46122A VNA (vector network analyser) to provide tangible results, referred to here as the small-scale proof of concept (SSPOC). This is necessary as it allows the group to test the available antennas, create a user interface and produce an automated detection system, all without being bottlenecked by the lack of a fully functioning GPR and not constrained to a specific application frequency range. Therefore, the second part of the GPR hardware is to create a specialised functional GPR prototype built by the team. This system is intended for field use, limited to the application frequency range and for the purpose of being attached to an appropriate drone, referred to here as the bespoke GPR prototype. The application frequency range is 450-750 MHz and is appropriate for the required mediums and size of features. Specific success criteria for this prototype in order to add value includes limiting the total system weight to 5 kg, and ensuring the GPR subsystem operates on a maximum power supply of 12V direct current. The success criteria of the SSPOC included achieving a minimum machine learning detection accuracy of 75% for detecting crevasses in a physical model. These specifications were set to guarantee the feasibility of future real-world deployments.

Building the bespoke GPR prototype involved designing a high frequency circuit using pre-built modules, designing low frequency processing circuits, programming a microcontroller to run the GPR and writing another programme to display the results. Building the high frequency circuits from basic components could have reduced the cost and weight of the system but there would be more potential for interference. The low frequency circuits distributed power to the bespoke GPR prototype systems, filtered the output signals from the high frequency circuits and amplified the necessary signals. The microcontroller was responsible for processing the GPR data, sending the processed data to a computer and for controlling the frequencies that would be produced by the high frequency circuits. The GPR prototype system could detect certain objects, but more time and resources would be required to get the system to produce results that could reliably detect for Arctic crevasses.

Signal processing algorithms were developed and tested to transform raw radar data into clearly interpretable radargrams (A-scans and B-scans), involving filtering, gain adjustment, and normalisation steps. Finally, a complete and thorough graphical user interface (GUI) to display these processed radargrams and for a user to interface with, which would also have automated detection capabilities.

Automated detection involved training and validating an object detection model to automate the identification of crevasses in radargram images. The YOLOv8 framework was selected for its proven effectiveness in real-time applications. The model was trained using a custom dataset of radargrams with carefully annotated bounding boxes.

To facilitate accurate and repeatable testing, a laboratory-based test rig was designed and constructed. To mimic the relative motion between a low flying drone and a polar environment, a rig was implemented which uses a stepper motor-driven moving medium beneath a static GPR system. This configuration ensured the consistency, reliability, and quality of the radar data acquired. The synthetic testing medium had dimensions of 1.2 metre by 1 metre by 0.4 metres. It comprised layered expanded polystyrene foam with a central clay core and was developed to mimic the dielectric contrast of a real crevasse, ensuring that radar signals closely resembled those expected from field conditions.

Antenna performance was evaluated through tests; different antennae provided different advantages. Vivaldi antennae could emit radio signals in a specific direction, and they worked with a wide frequency range[2]. Alternative specialist V14 antennae produced successful detections at short ranges and could also work with a wide range of frequencies. Several crosstalk, frequency range and object detection experiments were conducted and documented, resulting in building a frame for the Vivaldi antennas that held them securely at 20 cm apart, used radar absorbing material and appropriate noise reduction techniques.

The development of a robust GUI was another critical component of the project. The GUI was structured into modules and written in python providing users with intuitive UAV control through joystick input, real-time visualisation of GPR data, and system diagnostics via a clearly designed dashboard. Additional modules focused on radargram post-processing and crevasse detection. To achieve this, the GUI integrated the trained YOLO model, enabling users to easily load radargram images, execute detection algorithms, and export annotated images. The interface was demonstrated successfully in controlled simulations, establishing its usability and functionality.

Hence the project successfully demonstrated a lightweight, functional GPR prototype to be built capable of producing radar data. Furthermore, a complete proof of concept was executed and tested using an Anritsu VNA and documented. Visualisation of radar data was achieved via instant generation of both A-scan and B-scan plots, enhancing immediate interpretability during scans. The trained YOLO model met required accuracy targets, providing robust and reliable automated crevasse detection. Additionally, the laboratory test rig allowed for consistent and controlled radar data collection, significantly enhancing testing validity. Finally, the integrated GUI demonstrated a practical combination of UAV control, radar data management, and detection analysis, laying groundwork for future field applications.

Several key areas remain to be addressed in future development. The transition from laboratory validation to real-world field deployment on an operational UAV represents the next significant step. Further hardware miniaturisation and product design is necessary to achieve improved UAV endurance and ease of transport in harsh environments. Real-time onboard processing and detection during UAV flights should be prioritised to eliminate post-processing delays and enhance mission efficiency. Lastly, expanding the radargram dataset to cover diverse environmental scenarios and subsurface conditions will improve model generalisation and accuracy, ensuring reliable detection across varied real-world deployments.

Overall, the project has successfully established the foundation for a viable, UAV-based GPR solution for crevasse detection. It has demonstrated proof of concept through rigorous laboratory experimentation, simulation, and validation, meeting or exceeding all established technical targets and laying clear groundwork for continued development towards practical deployment.

1. Introduction and Project Vision

1.1. Project Motivation

Crevasses are deep, mostly concealed cracks in glacial ice. They pose serious threats to the safety of explorers, researchers and other teams (e.g. emergency responders) that operate in polar regions. The crevasses can reach depths of tens of meters and widths exceeding several meters, with their density and orientation varying seasonally and geographically. A partially hidden crevasse can be seen in Fig 1. A study conducted in Greenland found that crevasses could reach depths of up to 25 m, with significantly different dimensions across the glacier surface [3].

In winter, all crevasses are obscured by snow bridges, thin layers of snow and compacted ice formed by wind and precipitation. These structures render visual identification nearly impossible and significantly increase the danger of accidental falls. They tend to be more prevalent and deceptive in the colder months, making crevasse detection even harder during winter operations [3].



Fig 1. *A sinister Antarctic crevasse lurks under a snow bridge [4]*

Traditional methods of detection, such as manual probing or exploration by using tethering ropes, are inefficient, slow, and hazardous. The motivation for this project arises from the emergent need for a safer, more accessible, cost-effective method to identify crevasses in remote environments, even if they are hidden. Unmanned Aerial Vehicles (UAVs) provide an ideal platform thanks to their mobility, flexibility, and ability to reach inaccessible terrains in polar regions. By integrating a GPR system into a UAV platform, the project proposes a scalable solution that can significantly improve awareness during Arctic and Antarctic expeditions, disaster rescue operations, and remote surveying missions.

Example applications of the project are:

- Scientific research expeditions over glaciers.
- Polar search and rescue missions.
- Pathfinding for robotic rovers or autonomous platforms.
- Remote geological surveying.
- Avalanche risk assessment.

1.2. Aim and Objectives

This project aims to develop a lightweight, UAV-mountable Ground Penetrating Radar (GPR) system to detect visually obscured crevasses in polar regions. The system integrates radar signal acquisition, signal processing, and machine learning to interpret sub-surface cavities. The GPR system is designed for efficient subsurface imaging while being user-friendly and cost-effective. The objectives are as follows:

- Conduct a small-scale proof of concept with complete test results with an existing VNA, complete with a repeatable, robust testing rig.
- Develop a signal processing pipeline for radar data to produce clearly interpretable and industry standard radargrams.
- Implement a machine learning algorithm for automated analysis and achieve over 75% accuracy.
- Develop a compact, bespoke GPR prototype with specifications suitable for deployment on a UAV.

1.3. State of the Art Review

A wide range of techniques have been explored in recent decades for crevasse detection. Each one offers different trade-offs in terms of cost, resolution, portability, and suitability for real-time deployment in remote and hazardous terrains. To define the technical scope of this project, a critical review of current technologies was conducted and evaluated their feasibility for lightweight UAV-based deployment. Existing crevasse detection methods include:

Yeti Robotic Platform (NASA)

The Yeti rover, developed as a part of NASA's Operation "IceBridge", is a remote-controlled ground robot that weighs approximately 81 kg [5]. It was designed to move a radar antenna array across a snowy terrain to map voids and crevasses, as can be seen in Fig 2 (left). Even though Yeti improved safety by eliminating the need for human presence on unstable ice, its large size and heavy weight limited its operational flexibility. It was firstly used for scouting paths for heavy vehicles rather than personnel. In narrow snow bridges or sloped surfaces, Yeti still faced the same risks of falling through the ice like a human would.



Fig 2. Yeti Robotic Platform (NASA) (left) and Helicopter-mounted GPR (right) [5], [6]

Helicopter-mounted GPR

Another experimented approach is the deployment of GPR systems using helicopters as seen in Fig 2 (right) [7]. These platforms were able to carry large, high-resolution radar arrays by scanning broad areas from low altitudes. However, helicopters were expensive to operate. They require specialized pilots, significant infrastructure and can pose safety risks especially in adverse

weather. Additionally, their use is typically limited to large-scale missions supported by national or scientific institutions.

High-altitude Synthetic Aperture Radar (SAR)

Spaceborne and aircraft-mounted Synthetic Aperture Radar (SAR) offered a high-level overview of glaciers over a large area [8]. SAR system achieved a resolution by simulating a large antenna aperture using platform movement. This narrowed the beam width. However, the detection depth and resolution of SAR remained insufficient for identifying narrower crevasses that could trap a human. Additionally, SAR lacked real-time operation capability, which is crucial for teams navigating hazardous regions. Refer to Fig 3 (left). Furthermore, this method still needs to be verified.

Manual Radar Towing (Ground-based GPR)

Portable ground-based GPR systems such as the ones used in construction and utility mapping are sometimes manually used by researchers across glacial surfaces [9]. These systems are more accessible and cost-effective. However, they are physically demanding and limited by the navigability of the terrain. They are slower for long distances and pose safety risks when operated over concealed snow bridges. Refer to Fig 3 (middle).



Fig 3. *High-altitude Synthetic Aperture Radar (SAR) (left) [10], Manual Radar Towing (middle) [11] and Existing GPR UAVs (right) [12]*

Existing GPR UAVs

There were several commercially available UAVs with GPR capabilities [12], [13]. These systems were usually designed for various purposes, although most could be used to detect the thickness of snow/ice. Using GPR to detect snow/ice can be effective in perfect conditions. However, the performance of the GPR could vary significantly depending on the frequencies used. It is important to note that these systems were not designed specifically for detecting polar crevasses, unlike this project, which focuses on this from the beginning. Refer to Fig 3 (right).

Positioning This Project's Approach

This project introduces an approach that balances the limitations of the existing technologies mentioned. By mounting a Ground Penetrating Radar system on a lightweight UAV, the solution enables high-resolution scanning at low altitudes that is both portable and cost-effective. The platform is designed to operate closer to the surface than satellite-based systems. This improves resolution while reducing helicopter or rover usage's logistical complexity and expense. Facilitating machine learning enables real-time interpretation of radargrams. This configuration is particularly suitable for hazard detection in remote, ice-covered regions where human access is limited and safety is especially important.

1.4. Project Overview

This project develops a compact GPR system designed for integration with a UAV, specifically aimed at detecting hidden crevasses in polar environments. Due to limitations in testing, the delivered system separately delivers a scaled-down proof of concept and a specialised bespoke

GPR prototype rather than a fully deployable solution. It, however, still successfully demonstrates essential functionalities, including radar signal generation, data acquisition, signal processing, and automated crevasse detection via machine learning. The project consists of two main phases:

- **A Lab-based Testing Platform:** A controlled setup enabling validation of the signal processing pipeline and machine learning algorithms using radar data from a Vector Network Analyser (VNA).
- **UAV-mountable Prototype:** Custom-designed GPR hardware developed for UAV integration, emphasising weight minimisation and power efficiency.

The overall testing rig including all aspects of the project, such as the Anritsu VNA, the Bespoke GPR prototype, crevasse medium, spooler as well as the actual testing rig and laptop for user interface can all be seen in Fig 4.

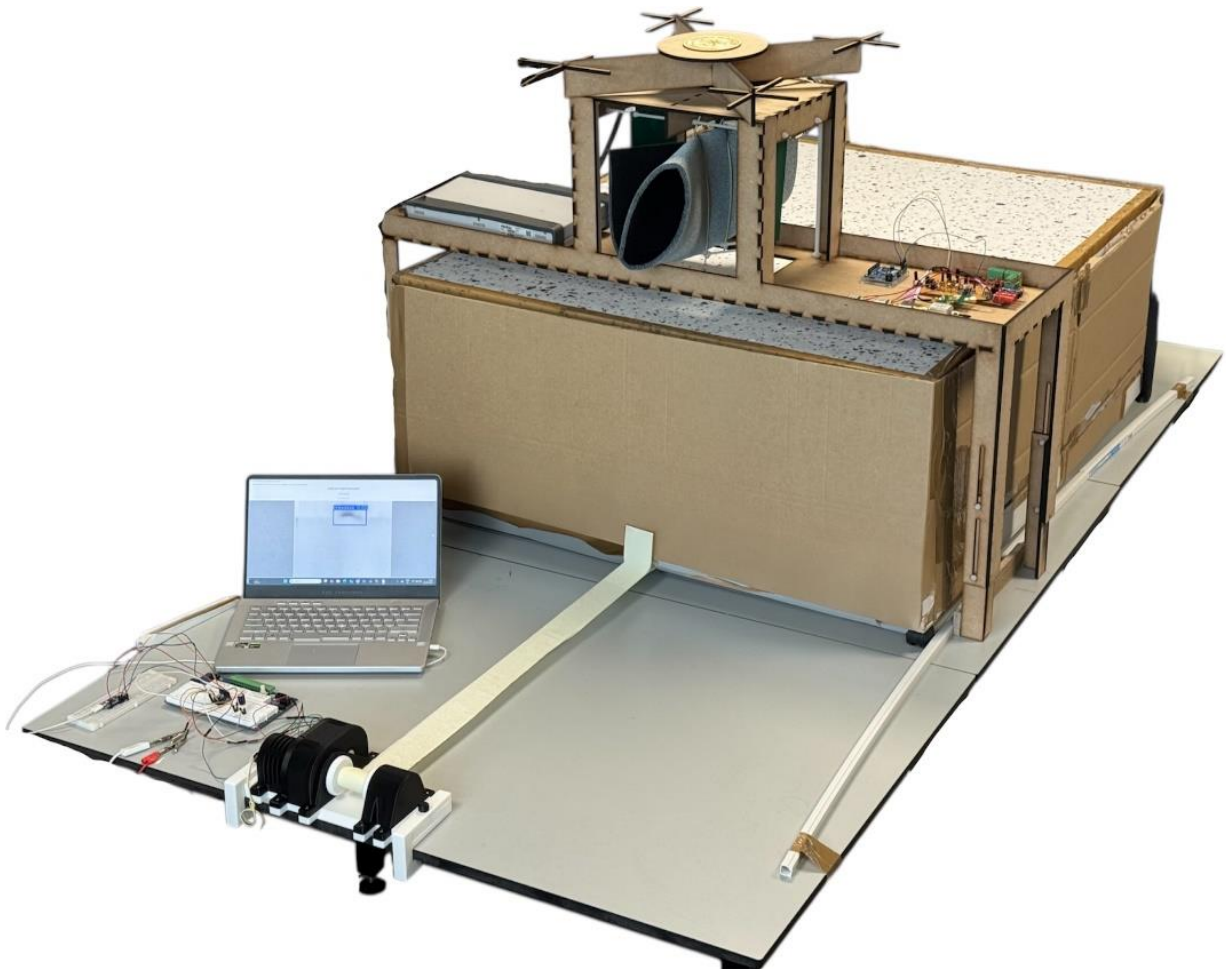


Fig 4. *Final project testing Rig with all subsystems included.*

A system-level block diagram provided in Fig 5 illustrates the architecture and integration of components across the project scope including GPR, signal processing, UAV platform, and user interface components. Subsystem-specific details will be discussed in later sections, along with testing methodologies, challenges encountered, and achievements to date. The interrelation among the key subsystems is as follows:

- **GPR Hardware:** Generates radar signals, transmits them into the environment, and receives reflected signals from subsurface anomalies.
- **Signal Processing:** Converts the received radar signals into clear radargrams through filtering, Fourier transforms, and normalisation.
- **Machine Learning Detection:** Processes radargrams using a trained YOLO-based model to identify crevasses automatically.

- **Graphic User Interface (GUI):** Integrates data visualisation, system controls, and displays detection results, facilitating user interaction with all components.
- **UAV Simulation and Test Rig:** Provides a demonstration of simulated drone control. Includes building an appropriate test rig to simulate a low flying UAV.

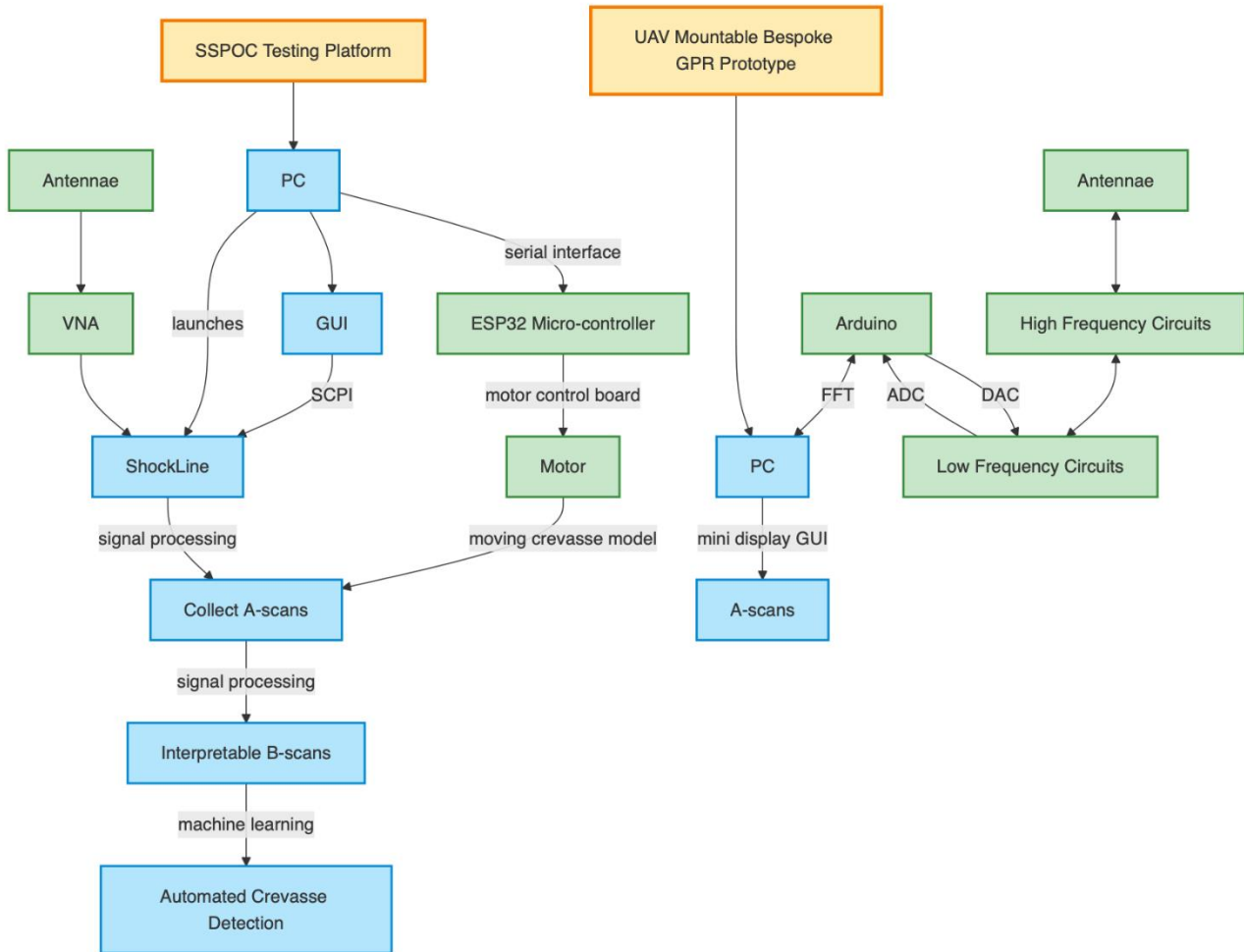


Fig 5. System-level architecture diagram illustrating the integration of major subsystems.

These subsystems are closely integrated through communication protocols, facilitating modularity and future scalability. The system-level block diagram included in Fig 5 clearly illustrate how these components interact, identifying inherited technologies and newly developed elements. The 2 main overarching systems are shown in amber, hardware components in green, and software subsystems in blue. This approach enables thorough subsystem validation and demonstrates significant progress towards a practical, UAV-mounted crevasse detection solution for polar exploration and safety enhancement.

2. Technical Advancement & Results

2.1. Bespoke GPR Prototype

2.1.1. Overview

At the start of the project, the team set an ambitious goal of building a GPR system itself. Initial research was completed to investigate the feasibility of producing a prototype GPR within the project budget (£1,650). The research suggested that producing a low-cost GPR system within the budget was feasible [14].

Their method involved using a VCO (voltage-controlled oscillator) to produce a time-varying high-frequency signal, which was then split into two. One signal was transmitted through an antenna, then reflected off an object before being received by another antenna and combined in an RF (radio frequency) mixer with the other signal that came straight from the splitter. The difference in time between the two signals from the mixer and the constantly changing frequency resulted in a difference between the two signals being fed into the mixer. The mixer then multiplied the two signals before going through a low-pass filter. This resulted in a low-frequency signal, whose characteristics are related to the distance of the object being detected by the GPR. To extract these characteristics, the output signal is then passed through an FFT (Fast Fourier Transform) to result in a GPR system capable of producing meaningful outputs.

To ensure the core concepts would work, a basic simulation was completed using Simulink. The simulation (seen in Fig. 6) generated a frequency-modulated sine wave before splitting the signal into two. Both paths were connected to a multiplier, but one was delayed. The output of the multiplier went through a low-pass filter before displaying the signal's frequency components.

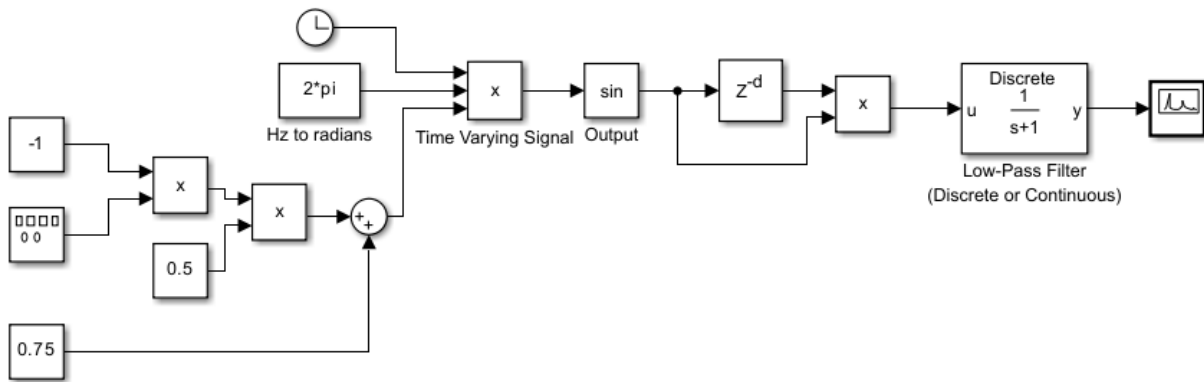


Fig 6. Simulink model

As the delay increases, the frequency component of this delay shifts along the frequency domain to the higher frequencies, decreasing in magnitude (see Fig 7). These results were deemed acceptable as the bespoke GPR prototype was intended as a proof of concept for this project, and signal processing could be added to amplify higher frequency components to reduce the difference between the signals for each delay.

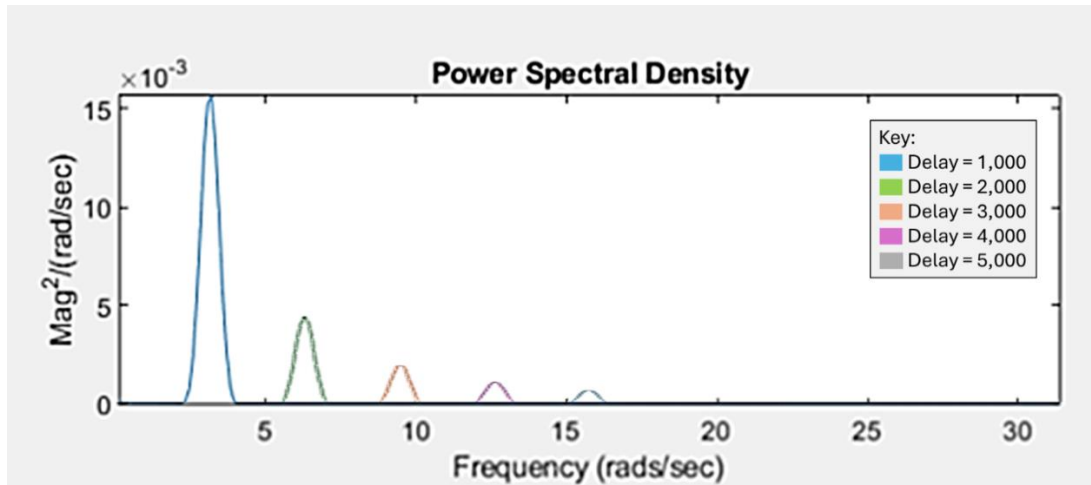


Fig 7. Simulink simulation results with a delay of 1,000 to 5,000 (in steps of 1,000)

Once the simulations were complete and showed that the GPR would theoretically work, calculations were completed to show the expected response from the GPR. The following equations were taken from [14], where (1) shows that the time, t , for a signal to be emitted from an

antenna and then received by the other antenna is related to the distance of the object detected by the GPR (D).

$$t = \frac{2 \cdot D}{c} \quad (1)$$

$$f_{out} = f_1 - f_2 = \Delta f \cdot t \quad (2)$$

The output frequency from the mixer (f_{out}) is the difference between the two input frequencies (f_1 and f_2), seen in (2). It also shows that the output frequency is related to the frequency change of the VCO output (Δf) and the time between the signals.

(3) shows that the rate of change of the frequency is related to the maximum frequency used by the VCO (f_{max}), the minimum frequency used by the VCO (f_{min}) and the time of a complete cycle of the modulation frequency (T).

$$\Delta f = \frac{2 \cdot (f_{max} - f_{min})}{T} \quad (3)$$

$$f_{out} = \frac{4 \cdot D \cdot f_{mod} \cdot (f_{max} - f_{min})}{c} \quad (4)$$

(1), (2) and (3) are combined to produce (4). It shows that the frequency at the output of the mixer depends on the modulation frequency, the range of frequencies being produced by the VCO, the speed at which radio waves travel and, importantly, the distance of the object being detected by the GPR.

(4) can be separated into the two equations (5) and (6). (5) contains variables that the GPR can control, while (6) shows that the frequency output of the mixer is proportional to the object's distance.

$$k = \frac{4 \cdot f_{mod} \cdot (f_{max} - f_{min})}{c} \quad (5)$$

$$f_{out} = k \cdot D \quad (6)$$

(6) agrees with the simulation results from Fig 7 as the magnitude of the signal is proportional to the time of the delay. Both the simulated results and theoretical equations justify the operation of the GPR.

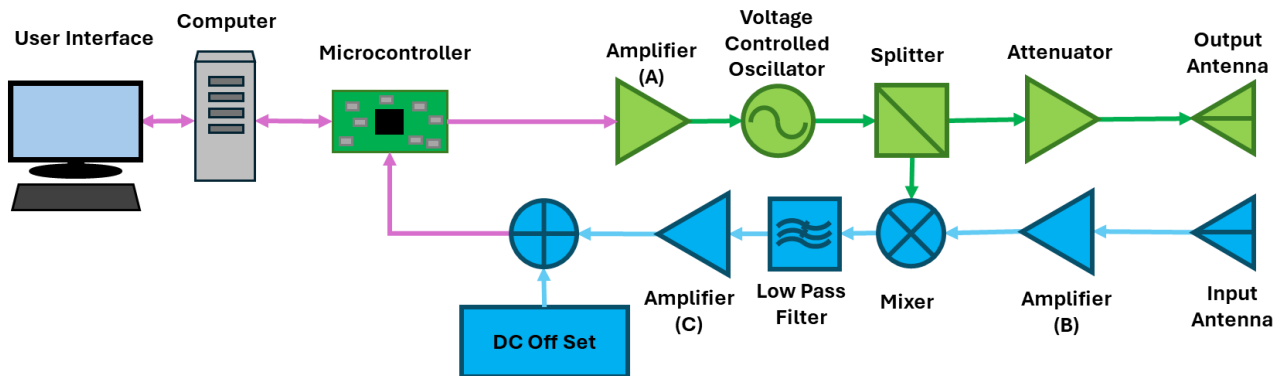


Fig 8. Bespoke GPR prototype block diagram

Research regarding related GPR systems allowed the group to create a diagram, shown in Fig 8, of a GPR system tailored to the needs of the project [15]. The design enabled a computer to present the results and control the GPR via a microcontroller. The microcontroller generates an amplified signal (amplifier A) to regulate the VCO across the full range of frequencies. The signal from the VCO is then divided into two, with one portion connecting directly to the mixer and the other portion being attenuated (to ensure Ofcom compliance) before being transmitted through an antenna. When an RF wave reflects off an object, it is received by the other antenna, amplified via amplifier B, and subsequently sent to the mixer. The output of the mixer is then lowpass filtered (to ensure that only the expected frequencies are received), amplified by amplifier C (to achieve a

peak-to-peak voltage of 5 V) and then a DC off set is added to ensure that the signals being sent to the microcontroller is in the range of 0 V to 5 V. The ADC on the microcontroller then interprets the signals and performs an FFT on them before sending the data to the computer to display the results.

It was decided to use prebuilt modules for the HF circuit to enhance reliability and allow key components to be easily swapped or changed, while minimising the effects of HF interference. The low-frequency circuits would be constructed using individual components, as HF interference was less of a concern for them. This enabled these circuits to be cheaper, utilise more precise parts, and have a more specific design tailored to the circuit requirements. Fig 9 provides an overview of the circuits.

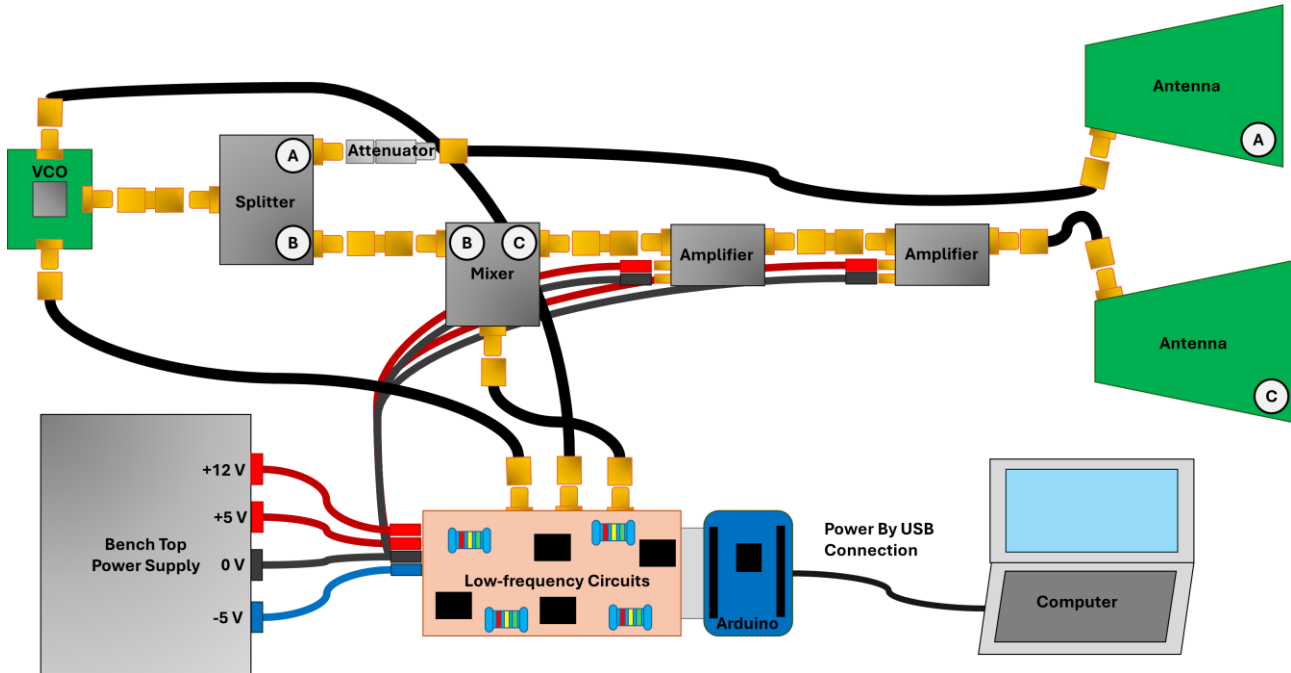


Fig 9. *Diagram of the bespoke GPR prototype*

The first component selected in the design was the VCO, which was responsible for the frequencies used by the GPR. Frequencies ranging from 400 to 900 MHz can penetrate through snow and ice, while a bandwidth between 300 and 400 MHz was determined as optimal [15] [16]. Based on these figures, it was decided that the bespoke GPR prototype should use a bandwidth of at least 300 MHz and a centre frequency of 600 MHz, enabling the GPR to detect features of ice, snow, and crevasses. A VCO specified to produce frequencies from 400 MHz to 800 MHz was used along with its development board.

For this project, it was decided to purchase a set of attenuators and amplifiers to provide flexibility in the design. Three attenuators were acquired for the project: two 20 dB attenuators and one variable step attenuator (0 dB to 31 dB). This setup allowed for a range of amplification from 40 dB to 71 dB attenuation, adjustable in steps of 1 dB. The amplifiers and attenuators were used to regulate the signal power supplied to the output antenna (A) and to ensure that the signals going into the mixer had the correct amplitudes. The output from the mixer was then directed into the low frequency circuits.

A crucial part of the low-frequency circuits was to serve as an interface between the microcontroller and the high-frequency circuits. This involved ensuring reliable power distribution, offering ESD (electrostatic discharge) protection, processing the signals from the high-frequency circuits for acceptance by the ADC, and adjusting the signals from the microcontroller to be compatible with the high-frequency circuits.

The microcontroller selected for the GPR was an Arduino Uno R4 Minima. It featured several necessary specifications, including a 48 MHz processing frequency, the capability for serial

communication with a computer, a 14-bit ADC (analogue to digital converter), and a 12-bit DAC (digital to analogue converter), all while remaining within the project budget [17].

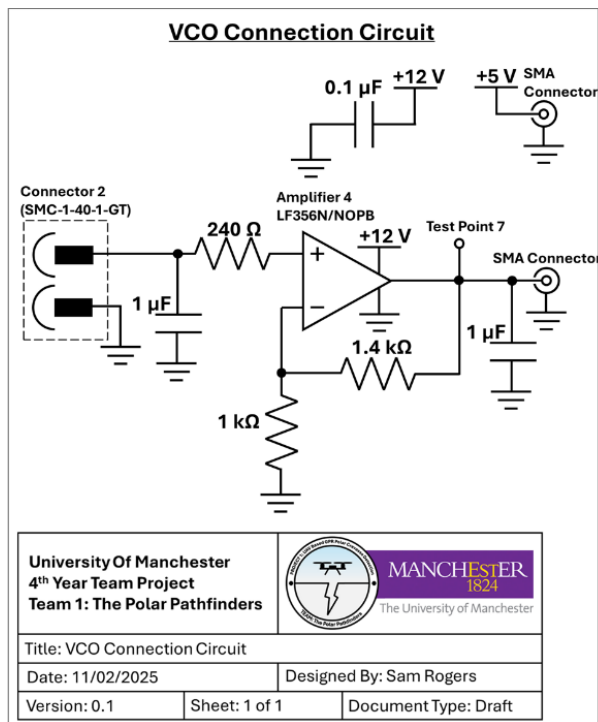


Fig 10. VCO Connection Circuit Diagram

The microcontroller's DAC was used to control the VCO. The DAC had an output range of 0 to 5 V, which was connected to an amplifier to allow it to give the 0 to 12 V signals that the VCO needed to operate. The circuit diagram for the amplifier can be seen in Fig 10. The output of the high frequency circuits (the output from the mixer) then needed to be processed before it could be connected to the ADC.

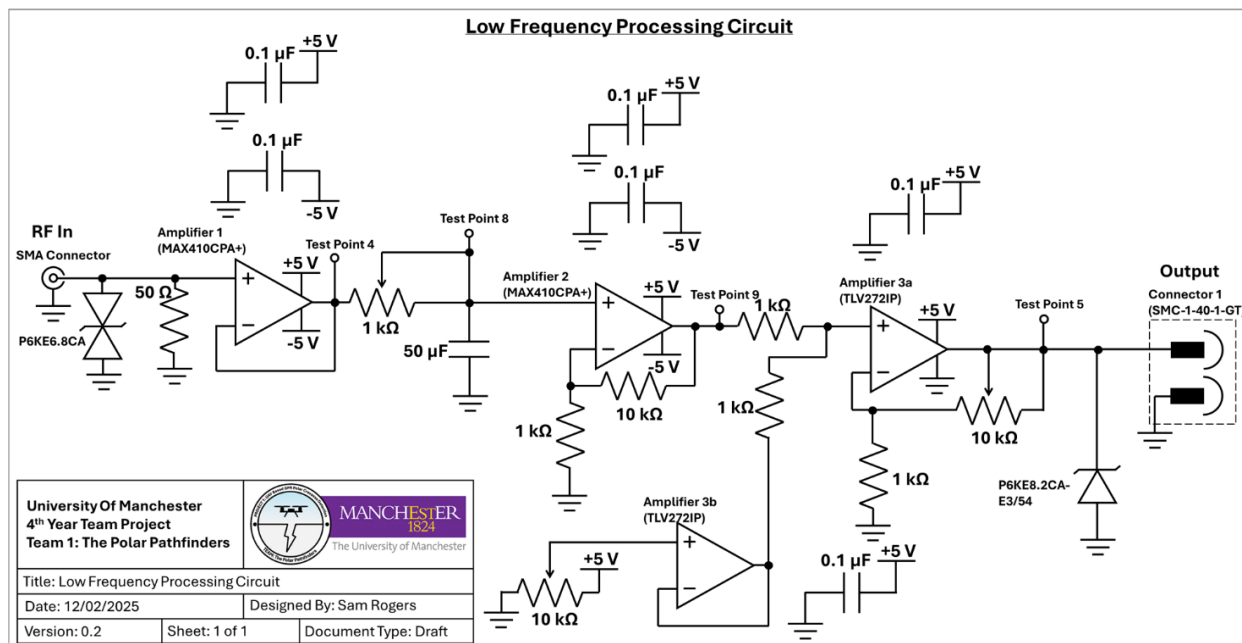


Fig 11. Low Frequency Processing Circuit Diagram

The signals from the mixer were inputted at the SMA connector labelled "RF In" and connected to a unity gain amplifier with a 50 Ω resistor to reduce the reflections seen in the HF circuits. Then the signals went through a low-pass filter (which can be adjusted to have a cut-off frequency from 3.2 Hz to 320 Hz) before being amplified as shown in Fig 11. These values were chosen as the

maximum frequency the mixer was likely to produce was 50.3 Hz, which showed an object's location at 1 m (calculated using (4), with $f_{mod} = 10$ Hz, $f_{min} = 400$ MHz, $f_{max} = 800$ MHz and $c = 3 \times 10^8$ m/s). A summing amplifier was then used with two potentiometers to add a voltage offset and to adjust the total amplification. It was important that the offset and the amplification could be adjusted to ensure that the signals going to the ADC were within the range of 0 to 5 V while having the largest amplitude possible. The low frequency circuits were used to interface with the variable step attenuator and to provide power to all the circuits. Fig 11 shows the schematics used for this.

2.1.2. System Output.

The output voltage of the real system can be seen in Fig 12, showing the output of the system, and validating it performs as expected.

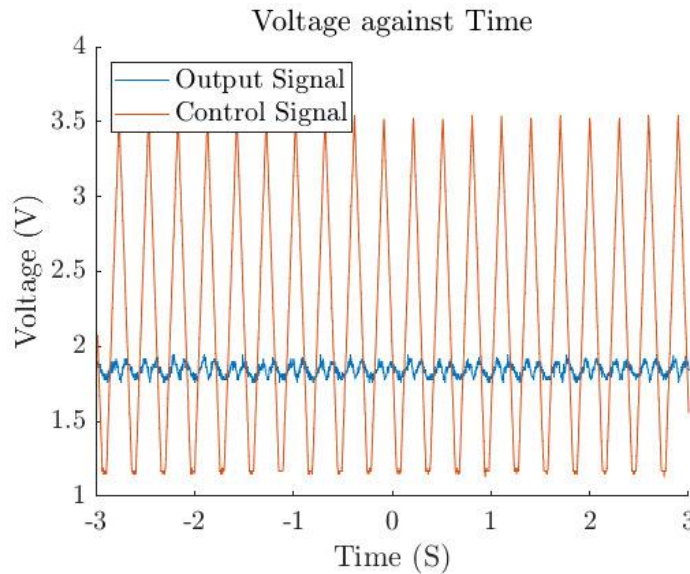


Fig 12. Output voltage viewed through an oscilloscope.

As shown in Fig 13, the attenuator connection circuit consisted of jumper connections to control the voltages on each of the attenuator connections. The power distribution circuit provided easy connections for each of the voltages required and used a converter to produce a 3 V supply. LEDs (light emitting diodes) were used to provide an easy confirmation that the circuits were receiving power, as shown in Fig 15.

To enable the hardware to run, firmware was created for the Arduino. The firmware was written using the Arduino IDE and was based around the ArduinoFFT example code [18]. Pin A0 on the Arduino had the ability to be used as a DAC (digital to analogue converter) and was set to produce a triangle wave as an output. Pin A5 was used as an analogue input and used the ADC to input the values from the signals processed using the low-frequency circuits. The analogue signal was then stored in an array and a FFT was computed using this data before the data was then transferred over a serial link to a computer.

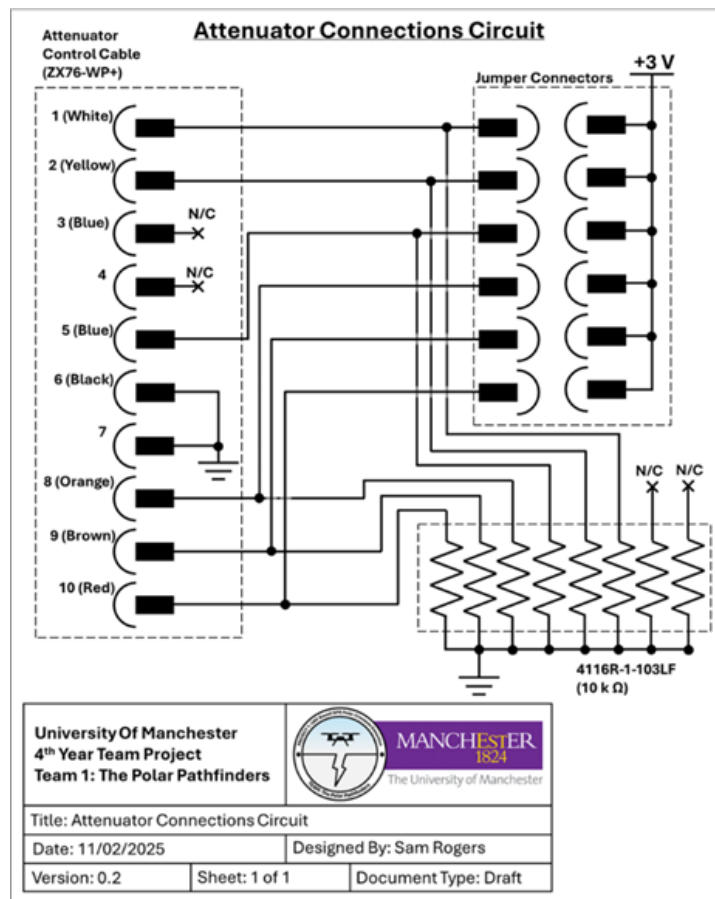


Fig 13.Attenuator Connection Circuit Diagram

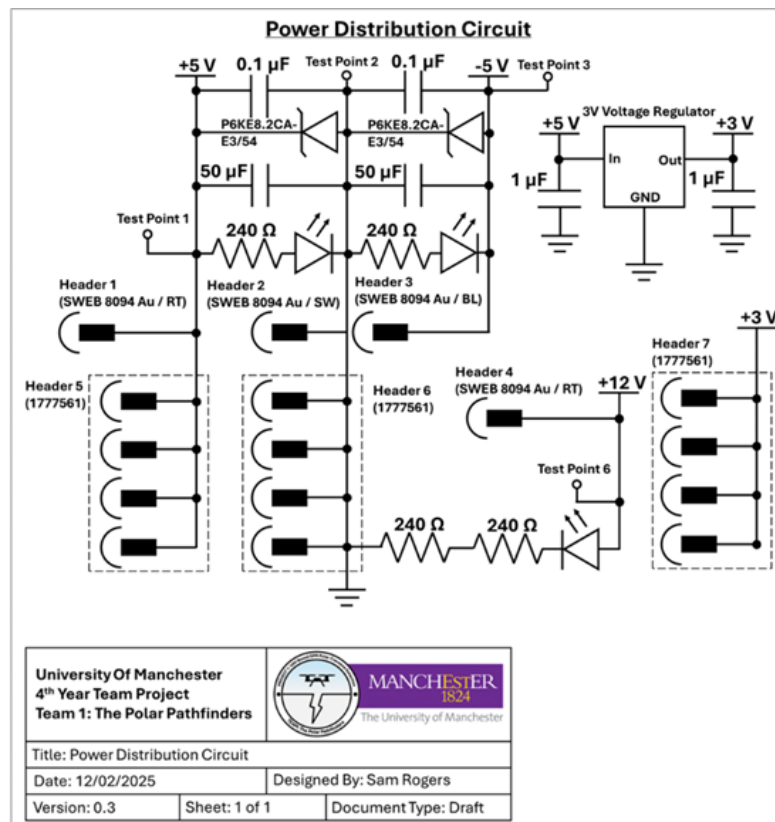


Fig 14.GPR Power Distribution Circuit Diagram

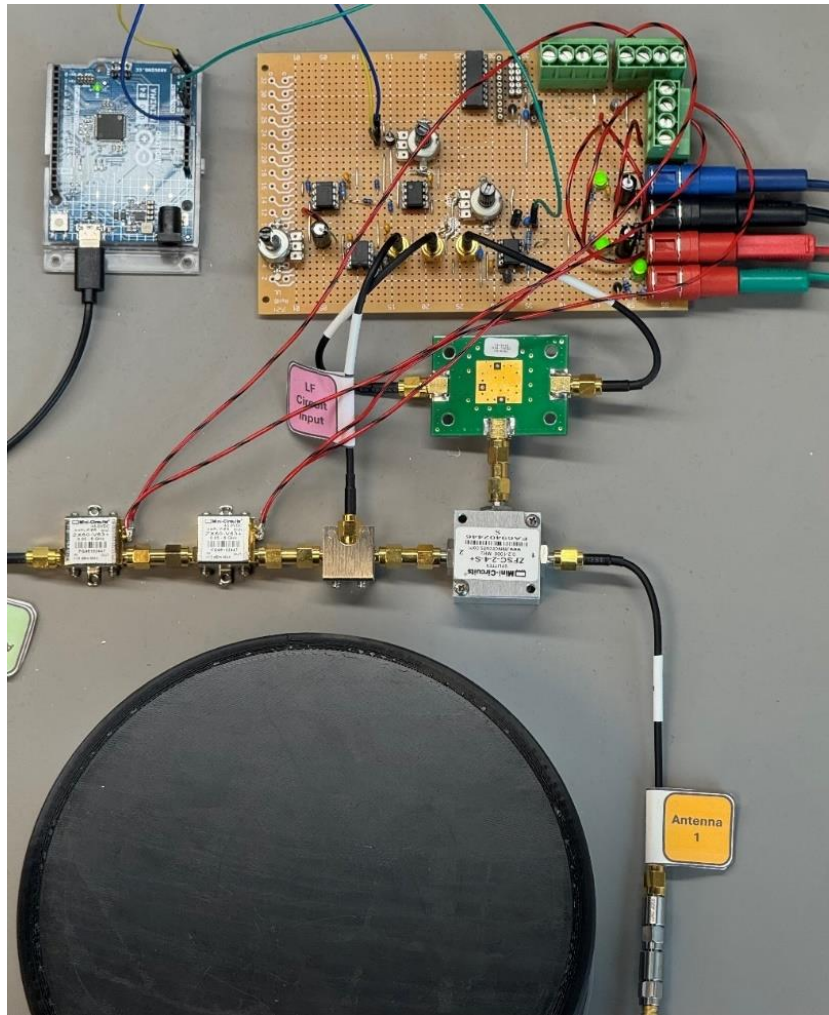


Fig 15. *Arduino GPR hardware*

Software

The computer programme used the data from the serial link and plotted it on a graph which refreshed every time a new set of data was received via the serial link. This software also allowed the data to be saved to a file.

Output

For the experiments using the GPR prototype, it was realised that the system was designed for detecting large crevasses through snow and ice, at relatively large distances. This meant that a successful result criterion at this stage of the project was simply to show a clear difference in results when the antennas had nothing in front of them and when they had a large object in front of them.

To comply with Ofcom's power regulations, an attenuator was required to reduce the antenna output power. The required attenuation was approximately -10 dBm to align with the legal limits. The placement of the attenuator is shown in Fig 9, and GPR power distribution in Fig 14.

It was decided to use V14 antennas and not the Vivaldi antennas for this experiment (the two types of antennas available during the project). V14 antennas were specially designed for landmine detection meaning that they had good performance at close ranges, which was needed for the GPR experiments as the power had to be limited while testing in the UK [19]. Furthermore, tests on the SSPOC indicated that the V14 were better suited for the task of crevasse detection (see Section 2.6). The data gathered by the GPR can be seen below.

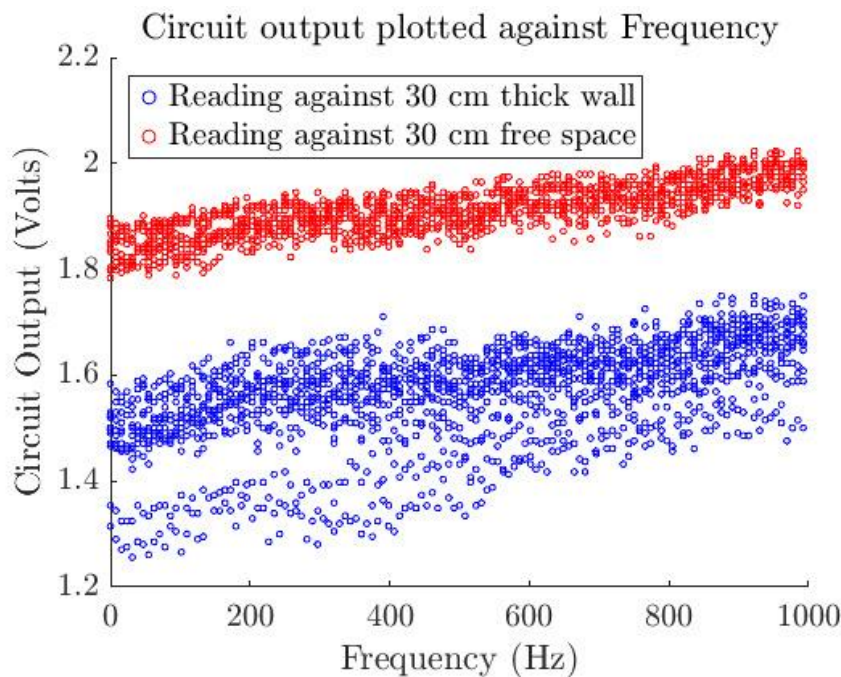


Fig 16. GPR Output

Testing The Bespoke GPR Prototype

The bespoke GPR Prototype, due to its designed frequency range, was theorised to only be capable of detecting larger testing mediums, which were out of scope for this project. However, it was found that large objects such as walls, larger sheets of metal and other sufficiently thick material and radar reflective mediums could be detected through the bespoke GPR system. However, due to a lack of space it was not feasible to perform the same tests as the SSPOC, as such a rig would need to be massive, modelling a real size arctic crevasse.

As seen in the Fig 16 above, there is a drop in the received signal intensity across all frequencies when the testing medium is changed. While scan resolution was hard to ascertain due to the imprecise nature of the testing mediums used, there is the definite ability to detect when a medium is present. It is notable, again, due to the frequency range selected, smaller objects which would normally be detectable for the SSPOC made no change in output for this system, this further confirms that the system is performing as intended: detecting larger voids while ignoring smaller gaps.

2.1.3. Antenna Choice

Overview

The selection of suitable antennas was critical for two primary functions in the SSPOC system: for transmitting and receiving GPR signals, and for providing the wireless communication between the UAV platform and the ground station, for future use in a real UAV-mounted system rather than the prototype. The antennas needed to fulfil several challenging requirements, which included a lightweight construction for UAV integration, compact size, sufficient bandwidth, minimization of self-coupling and minimization of environment interference to enable accurate crevasses detection.

GPR Antenna Design Considerations

- **Operating Frequency Selection:**

The choice of the GPR antennas was driven by a trade-off between penetration depth and resolution. Lower frequencies (less than 1 GHz) provide greater surface penetration, which is an advantage for detecting deep crevasses, and higher frequencies (up to 2–3 GHz) enable finer spatial resolution but with a reduced penetration. The testing phase deduced that the optimal range was 20 MHz–3 GHz (see Section 2.6).

- **GPR Antenna Types Evaluated:**

Several antenna designs were assessed regarding suitability:

- 1 **Bow-Tie Antennas:**

Bow-tie antennas offer a compact, broadband structure, which makes them suitable for UAV mounting. Their small physical size and relatively omnidirectional patterns make them a good choice for lightweight platforms where antenna separation distance is limited.

- 2 **Vivaldi Antennas:**

Vivaldi antennas provide directional radiation patterns with higher gain across wide bandwidths, improving the signal-to-noise ratio for GPR systems. However, the ones used in testing are larger, with individual dimensions of approximately 15 cm × 22 cm, which introduces challenges for integration on a UAV with strict size constraints.

- 3 **Horn Antennas:**

Horn antennas have higher efficiency and gain but are typically bulky and heavy, making them unsuitable for compact UAV-based deployments.

- 4 **V14 Antennas:**

V14 antennas were designed for GPR systems focused on landmine detection [19]. They had several benefits, including being relatively lightweight, having low crosstalk and using a wide frequency range. However, these antennas are short-range.

Selected GPR Antenna Justification

Vivaldi and V14's antennas were selected for initial tests due to their gain advantage and directional properties. They were expected to enhance signal strength when acquiring radargrams. In the laboratory setup, the transmitting and receiving Vivaldi antennas were separated by approximately 20 cm. To reduce reflections caused by the environment, cross coupling and the noise induced from the platform, radar absorbing material (RAM) was placed between the antenna mounts.

However, subsequent analysis showed that a higher operating frequency range (2–3 GHz) requires a minimum far-field separation of approximately 65–97 cm to avoid significant near-field coupling. In the testing setup, operation was within the near-field region, resulting in partial coupling between antennae that affected the dynamic range and sensitivity of the radar. Even though RAM helped to mitigate environmental effects, it was not able to eliminate direct coupling between antennae at close range.

For the final bespoke GPR prototype, V14 antennas were selected (see Section 2.6). Their low cross-coupling, broad frequency coverage and compact profile made them more appropriate for the UAV-mountable system than the bulkier Vivaldi designs. While still not optimal for far-field separation on a compact drone frame, V14 antennas gave a balanced solution for short-range operation and compliance with size and weight constraints. Their application was informed by their proven use in landmine detection systems, which similarly require high precision in cluttered and near-surface environments.

2.1.4. Challenges and Considerations

Power Compliance

All RF elements were required to comply with Ofcom's power limits to ensure safe and legal operation. This was achieved by incorporating appropriate levels of attenuation at the output of the signal chain, which limited the transmitted power across the GPR operating bandwidth.

Signal Integrity

Addressing phase errors introduced by splitters and amplifiers through precise impedance matching. The signal chain included active and passive RF components such as splitters and

amplifiers. These can introduce phase shifts and reflections if they are not properly impedance matched. Careful matching was performed in the design to preserve the fidelity of the transmitted and received signals and improve radargram clarity.

Antenna Coupling and Signal Saturation

One of the challenges in a GPR system is the physical proximity of the transmitter and receiver antennas, which leads to coupling. Excessive coupling can result in a high-power direct signal path that interferes with the ability to detect weak reflections from subsurface features.

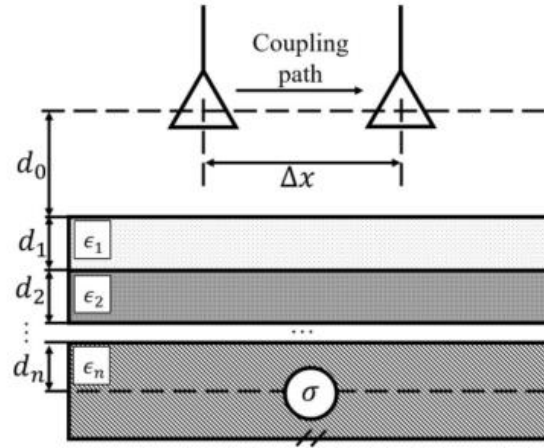


Fig 17. Coupling between transmitter and receiver antennas over a layered dielectric medium [20]

Fig. 17 above shows the coupling path between antennas and the multi-layered dielectric structure often seen in real-world glacial or subsurface environments. As shown in the figure, coupling occurs along a direct path between the antennas and can be especially problematic when operating in the near-field region. Two specific issues are associated with excessive coupling are:

- **Receiver Saturation:** Strong coupled signals may saturate the receiver's low-noise amplifiers (LNAs). This prevents the detection of weaker signals reflected from shallow crevasses.
- **Slope Removal Artifacts:** Attempting to remove a large, coupled signal slope during post-processing can introduce additional noise in the system and it reduces radargram quality.

To avoid amplifier compression and signal loss, it is essential to ensure that the instantaneous peak-to-peak voltage (V_{pp}) of the coupling signal remains below a critical threshold. This limit is defined by the (7) [21]:

$$V_{pp}^{max} = \frac{2 \cdot V_{sat}}{G_{amp}} \quad (7)$$

Where V_{sat} is the saturation voltage of the amplifier and G_{amp} is the gain applied at the receiver input.

2.2. Signal Processing Subsystem

2.2.1. Overview

The overall goal of the project's signal processing algorithms is to take the radar system's raw data and ensure that it is converted into a format that the user can easily read. These algorithms will do this by converting the raw data into an A-scan, representing how a single radar pulse changes over time due to various absorptions and reflections from changes of mediums. A B-scan combines multiple A-scans along a specific direction to provide a cross-sectional subsurface profile visually illustrated in Fig 18, where a parabola has been detected in a B-scan. This concept can be

extended further where multiple B-scans can be combined to create a 3D image of a subsurface, again shown in Fig 18. To ensure that these scans correctly identify hidden crevasses, steps such as applying window functions, background subtraction and inverse square magnitude, but these will both vary greatly when using the Anritsu MS46122A or the project's own GPR system.

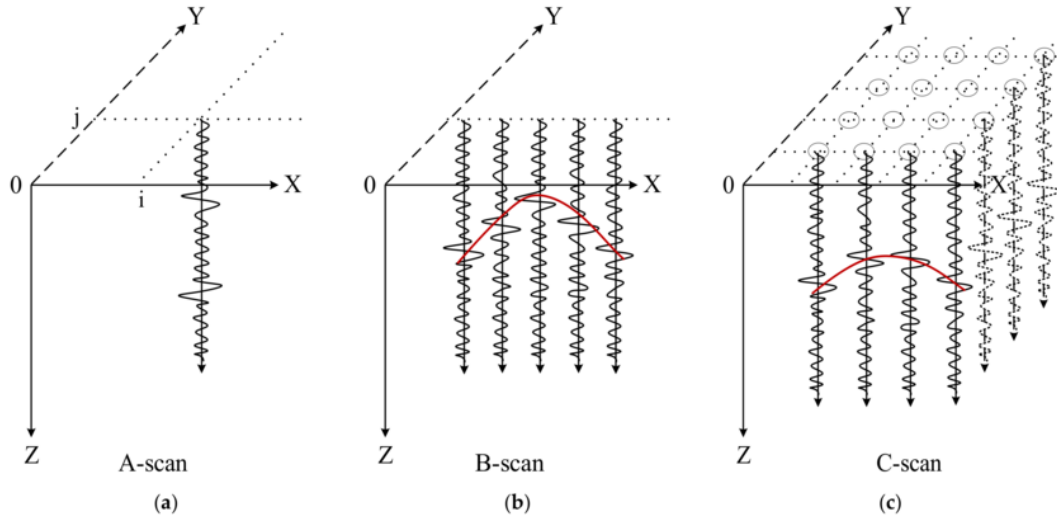


Fig 18. Visual representation of A-scan, B-scan and C-scan [22]

2.2.2. Data Acquisition by Using SCPI.

Standard Commands for Programmable Instruments (SCPI) is a communication programming language that facilitates sensor and collection system interfacing. SCPI was required to extract A-scans from the Anritsu VNA at fixed intervals. Hence, it was used to request data repeatedly and consistently through serial communication. When requested, a Python function will periodically save A-scans as sets of .csv files at a frequency of 4 Hz. This frequency is limited to this, as while the program can save files at 800 Hz, the VNA itself is limited in the time it takes to filter and process the signals, resulting in oversampling, i.e., repeated readings if exceeded. The CSV files are saved in folders with a specific time index in a base folder.

The SCPI code used was taken from the official Anritsu examples GitHub page [23] and then further modified using the information within the ShockLine VNA Programming Manual available at [24]. By default, the VNA displays information in the frequency domain, but also can also output time domain data. There were many other configuration adjustments done with SCPI, but these were mainly to define scan resolutions, improve readability and enable communication. A brief of code is shown in Fig 19.

```
self.shockline_socket.write(":CALC1:PAR:COUN 1")
self.shockline_socket.write(":CALC1:PAR1:DEF S21")
self.shockline_socket.write(":CALC1:TRAN:TIME:TYP BAND") #Time Domain
self.shockline_socket.write(":CALC1:TRAN:TIME:TRIP ROUND") #Trip time is round
self.shockline_socket.write(":CALC1:PAR1:FORM REAL") # Format is Real
```

Fig 19. An example code snippet of using SCPI.

In the context of the overall program and testing setup, SCPI extracts individual A-scans from the Anritsu VNA over a serial interface at 4 Hz, for a particular run of the demonstration, and puts them in a timestamped folder. Due to the fixed collection rate and fixed rotational speed of the stepper motor, uniform localisation is ensured. Upon selecting any given folder in the user interface, signal processing is performed on it to put all the contained A-scans together, perform any additional normalisation and enhancement and hence produce a B-scan output.

2.2.3. Signal Processing of the SSPOC

For the SSPOC to be completed, the group needed to create a signal processing framework in which model crevasses could be detected and visualised intuitively for users. Serving as a milestone to prove the usefulness of this technology, and, if successful, would be a method to produce more B-scans to train the crevasse detection machine learning algorithm.

Initially, evidence was required to demonstrate that the B-scans produced by the signal processing algorithm could display an object detected by the Anritsu device when antennae were connected. Frequency scans were taken of a metal object below the antenna at a frequency of 1 Hz for 30 seconds, allowing the metal object to be placed and taken away for specific samples to test the system. As shown below in Fig 20, the frequency scans were successfully arranged to form a plot confirming that an object was detected when it was placed under the antennae.

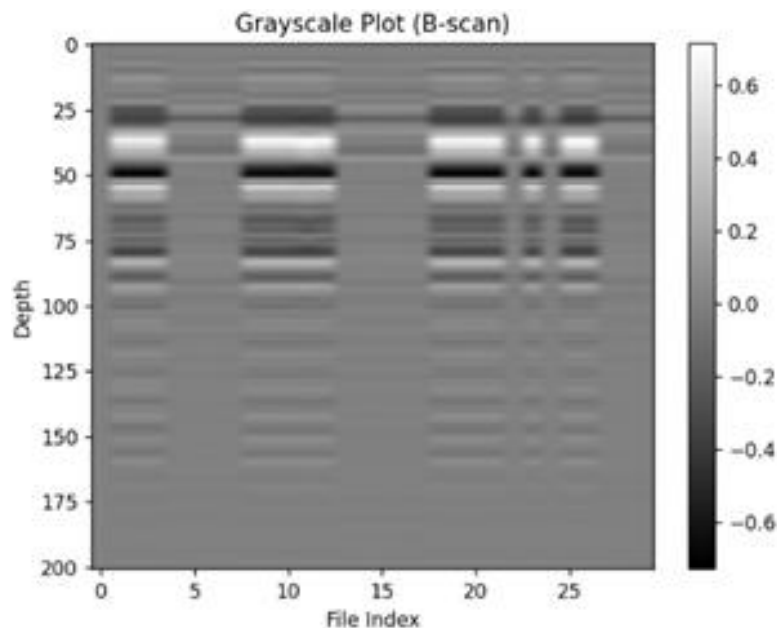


Fig 20. Frequency data of the Anritsu displayed in the same manner as a B-scan.

The data had background subtraction performed on it by taking the first frequency scan and subtracting it from all the subsequent scans [25]. Furthermore, if these frequency scans would be converted into the time domain, they would form A-scans, meaning that if the program in its current form would be able to plot valid B-scans. As the ShockLine Software used by the group to communicate with the Anritsu already contained functionality to convert the raw frequency data of the antennae into the time domain [26], the group's main work was to create a program that could seamlessly and quickly gain access to this data for processing, which is detailed in the previous section. By using the time domain data from the ShockLine software, the group were able to successfully display B-scan plots. Post processing features such as background subtraction and inverse square gain correction were implemented, but final control on which of these post-processing features would be used for the B-scans was given to the user as detailed in Section 2.5. A B-scan detecting a model crevasse is shown in Fig 21.

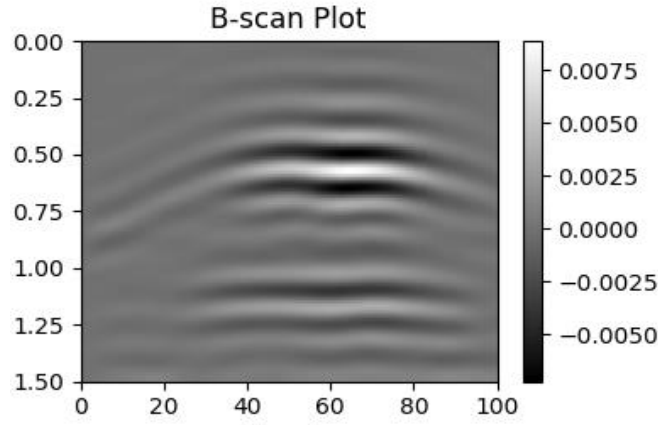


Fig 21. Example of B-scan from the final signal processing program of the VNA data.

2.2.4. Signal Processing of the Bespoke GPR Prototype

When using the Anritsu as a GPR system, many of the tasks were automated within the VNA itself and through the Shockline software. When the group began constructing its own GPR system, research was done to ensure that many of the processes done automatically in the VNA GPR system will successfully be implemented into the project's GPR. Therefore, this section will detail the steps that the group plans to take to produce a B-scan plot, coupled with the ranges of the detected targets.

The signal processing algorithm of the GPR prototype would receive beat frequency signals as its input. The beat frequency relates to the delay between the transmitted and received signals of the antennas, which will allow for the detection of targets, as each target will have its own beat frequency [27]. The representation of a singular detected target in equation form is shown in (8).

$$\text{Signal} = \sin(2 \cdot \pi \cdot f_{\text{beat}} \cdot t) \quad (8)$$

However, the signal that the signal processing algorithm will receive from the system will consist of far more frequencies, such as the one in Fig 22. The frequencies within this signal will contain beat frequencies of the detected targets; also included in the signal will be a clutter signal, and noise will be present. In the context of this project, the clutter signal will often be the ground's surface, which is different from a beat frequency signal, as a clutter signal may interfere with the detection of desired signals [28].

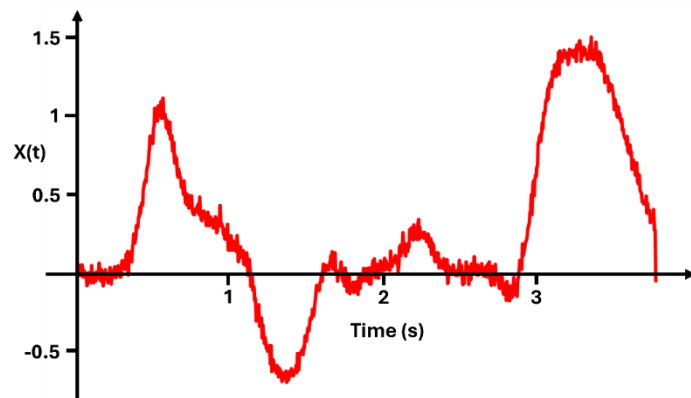


Fig 22. Example signal fed into the signal processing algorithm

Fig 21 is also represented in equation form in (9), which helps to summarise the signal's components.

$$\text{Signal} = \sin(2 \cdot \pi \cdot f_{\text{beat},1} \cdot t) + \sin(2 \cdot \pi \cdot f_{\text{beat},2} \cdot t) + \text{noise} + \sin(2 \cdot \pi \cdot f_{\text{clutter}} \cdot t) \quad (9)$$

To usefully process this data the signal will be run through an FFT over a specified chirp period, to allow for all the corresponding f_{beat} and $f_{clutter}$ frequencies to be identified, as shown in Fig 23.

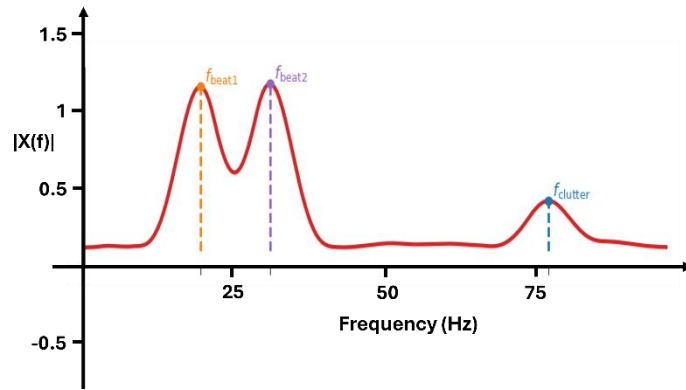


Fig 23. Frequency spectrum of Fig 22

While the FFT proves that the program has detected several targets, this does not yet have functionality to identify where these targets are in relation to the GPR system. Therefore, the range can be mapped by the f_{beat} of an object through (10) [29].

$$R = \frac{c \cdot T_c \cdot f_{beat}}{2 \cdot B} \quad (10)$$

Applying this equation to Fig 23 would result in an FFT range spectrum Fig 24 for example, providing a platform for the signal processing algorithm to create A-scans with identified ranges for detected objects. This would provide a straightforward way to produce B-scans.

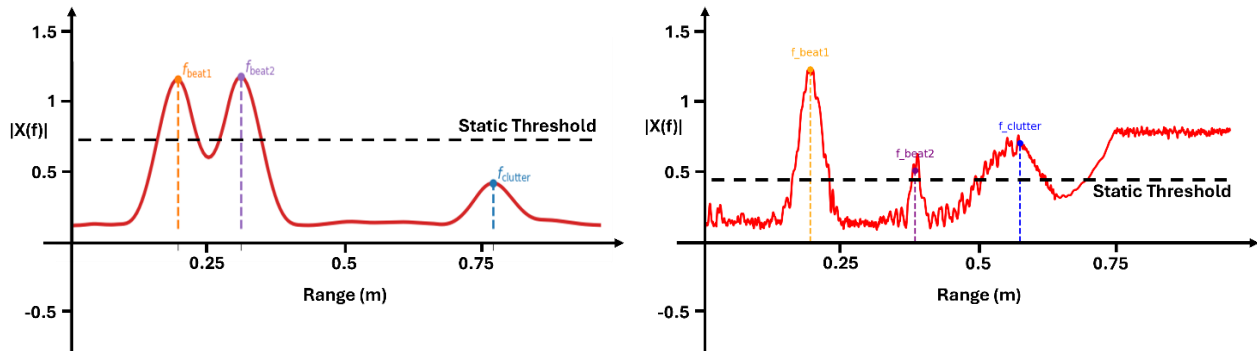


Fig 24. A simplistic FFT-range spectrum with a static threshold (left) and a more realistic FFT-range spectrum (right)

Typically, the output of the FFT-Range equation results in a spectrum in which several “false peaks” are within the plot, meaning a static threshold could not be applied to differentiate between detected targets and noise. In Fig 24 the higher frequency noise in the spectrum has a larger amplitude than the detected objects, showing the necessity for applying a non-static threshold to ensure that all detected objects are above the thresholding line, while filtering out false targets.

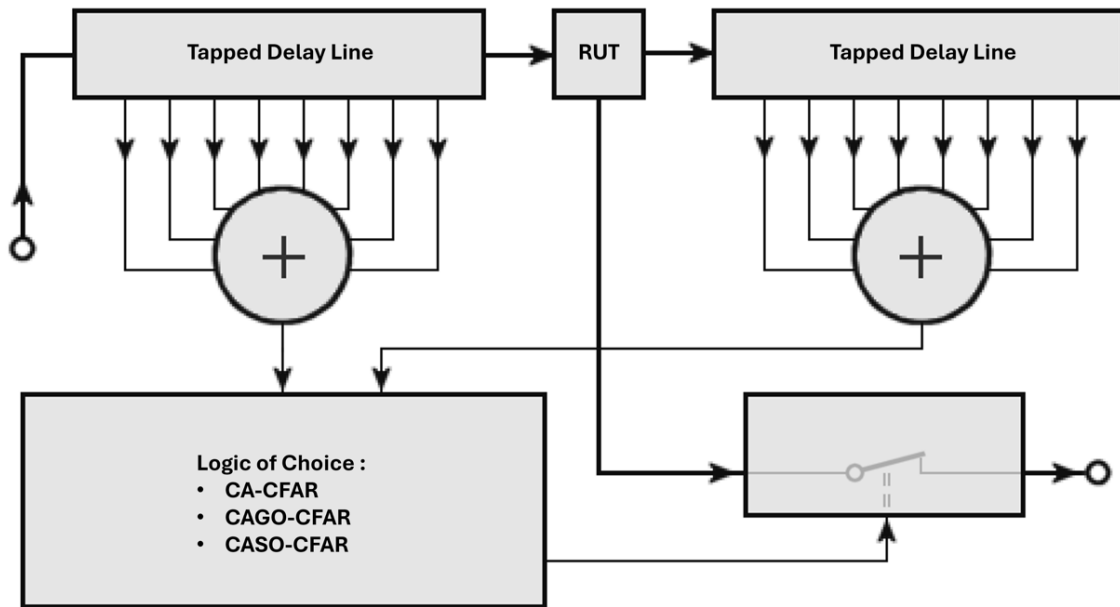


Fig 25. *Diagram of CFAR algorithm operation*

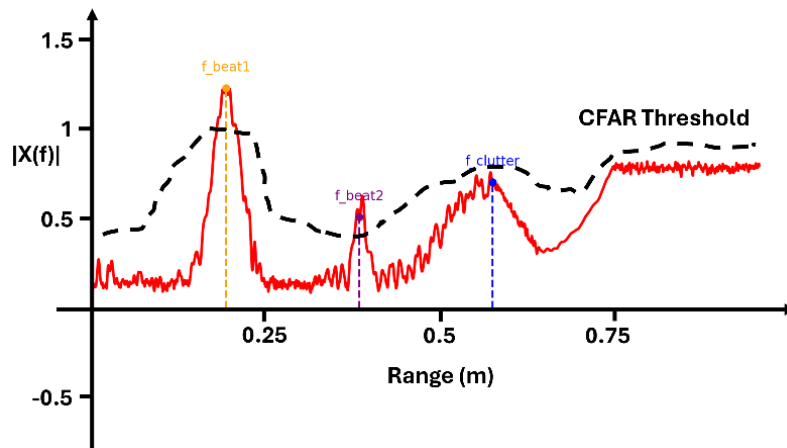


Fig 26. *FFT-range spectrum with CFAR Threshold*

One class of non-static thresholding algorithms are known as Constant False-Alarm Rate (CFAR). CFAR consists of two tapped delay lines which total and average all samples within the delay line, which both sandwich a range cell under test (RUT), shown in Fig 25. The RUT contains the element which will be tested against the threshold value, which will vary depending on the average values of the tapped delay lines. The RUT will also contain several gap elements chosen to correspond to the thickness of the peaks in the FFT range spectrum A-scans. Finally, a bias is applied to the threshold value, which in theory will result in a varying threshold value that will remove all noise, while keeping the detected objects' ranges like Fig 26 [30]. A summary of the various simple implementations of CFAR are shown in Table 1. However, it is worth noting that there exist more complex versions of CFAR, but these go beyond the scope of this project.

Table 1. Summary of all simple CFAR algorithms [31]

CFAR Type	Abbreviation	Explanation
Cell-Averaging	CA-CFAR	The average value of both tapped delay lines is used and a bias is applied to find the threshold value.
Cell-Averaging Greatest of	CAGO-CFAR	The tapped delay line with the higher average value has a bias applied to it to find the threshold.
Cell-Averaging Smallest of	CASO-CFAR	The tapped delay line with the smaller average value has a bias applied to it to find the threshold.

2.3. UAV Platform and Integration

2.3.1. Testing Rig and Setup

In lieu of a low-flying UAV, the early prototype of a testing rig comprised a platform mounted on four 40 cm tall PVC pipes with wheels capable of linear movement attached to the bottom of the legs. An encoder was to be attached to one of the wheels as a form of localisation for B-scan formation. An image of this can be seen in Fig 27.

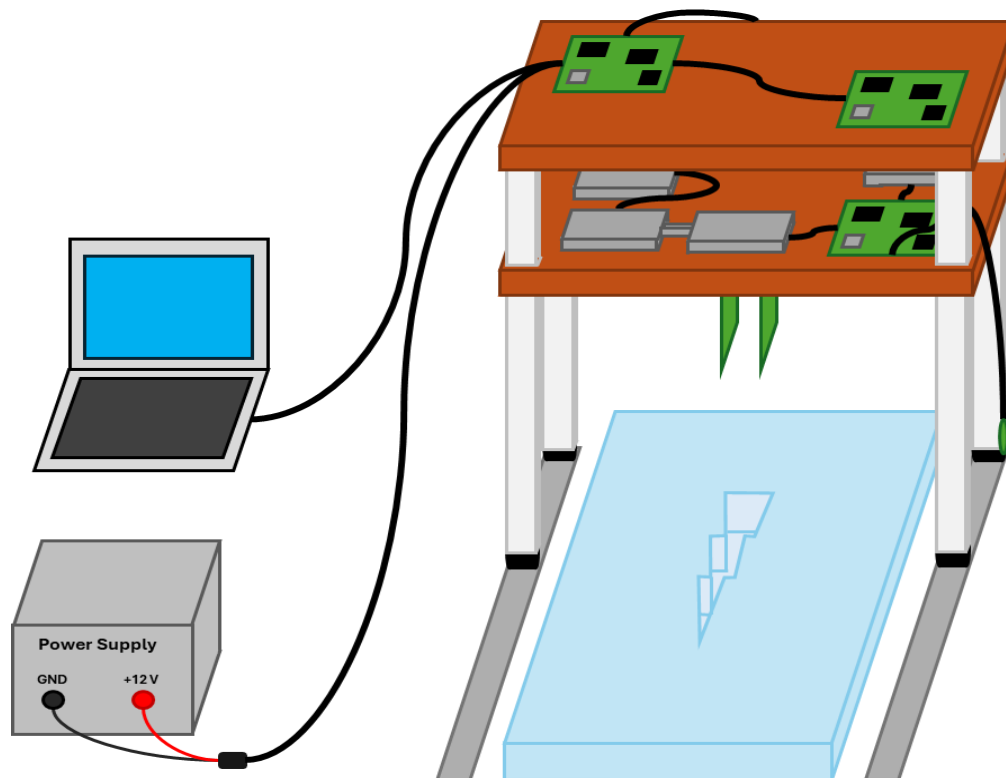


Fig 27. Early Testing rig to mimic low flying UAV (Virtual Model).

Disadvantages of this approach included the potential of the encoder causing interference in antennae, the introduction of significant additional noise characteristic of an indoor environment by a moving rig, and the fact that this noise would be dynamic and not be accounted for by background subtraction.

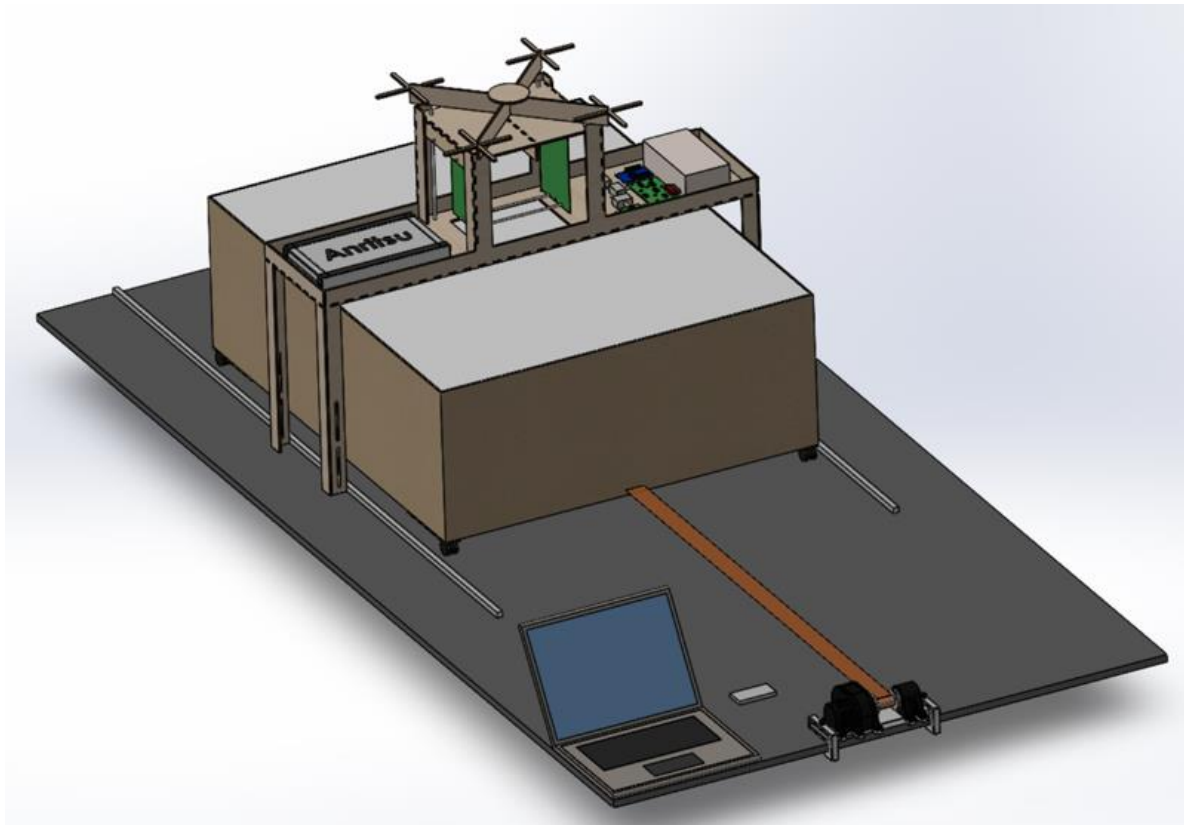


Fig 28. *Final Testing Rig to mimic low flying UAV (Virtual Model).*

Instead, a static rig and moving testing medium would capture the same relative motion for B-scan generation, with far viewer disadvantages, the only significant disadvantage being medium weight. A distant spooler powered by a stepper motor and gearbox for torque conversion pulls the medium under the static rig linearly. A complete comparison is shown in Table 2. A-scan collections are triggered for a set period of motor steps to ensure uniformly spaced and accurate B-scans, hence accounting for localisation as well. Visualisations of the updated rig can be seen in Fig 28. By substituting the localisation method with one suitable for a UAV, and incorporating appropriate anti-tilt mechanisms and stability, the setup can be translated to one appropriate for an actual UAV.

Table 2. *Comparison between mobile rig and static rig*

Feature	Mobile Rig	Static Rig with Moving Medium
Design Purpose	More accurate simulation of low-flying drone movement	Provided stable environment for radar acquisition with same relative motion
Localisation Method	Wheel-mounted rotary encoder	Motor step counting
Noise Level	High due to movement and indoor interference	Low, with consistent conditions which can be removed by background subtraction signal processing
Electromagnetic Interference	Potentially high from encoder directly under antennae	Minimal, motor placed away from antenna
Scan Consistency	Good but prone to oversampling if pushed too fast	High, due to fixed step-based triggering
Movement Mechanism	Manually pushed by hand	Linearly pulled by stepper motor

The aforementioned spooler for pulling the testing medium comprised a sophisticated 3D printed modular design with a gearbox that had replaceable gears, to allow for ratios of 1:1, 1:2.25, 1:4 and 1:6.25 to be used. It included a detachable pair of hooks to secure the winch to the edge of a table on which the test setup sits. Two different materials were used for the print. ABS (acrylonitrile butadiene styrene) was used for the parts that experienced the most mechanical stress (gears and spooler drum) while PLA (polylactic acid) was used for the housing of the winch. The gears were helical gears designed using a SolidWorks toolbox. The helical gear design was chosen to stop the gears from wearing in a single place and used a standard pressure angle of 20°. The winch held the motor, gears and spooler drum securely and had gaps to let the heat from the motor escape. The torque of the motor can be increased due to the gearbox.

$$T_2 = \eta \cdot T_1 \cdot \text{Gear Ratio} \quad (11)$$

The (11) shows how to convert torque by torque ratio. Where T_2 is output torque, T_1 is input torque, and η is efficiency.

$$F = \frac{T}{r \cdot \sin(\theta)} \quad (12)$$

The (12) shows how to convert torque to force. The force acts perpendicularly to the radius of the spooler, so θ equals 90 degrees and the sine term can be taken as 1. Here, r is equal to the spooler radius, which is 0.0125 m [32], and T is equal to 0.13 Nm. The final force output of the spooler with a 1:1 gearbox is hence 10.4 N.

The force required to pull the model, as measured by a newton meter, is approximately 8 N to overcome static friction and 6 N to overcome rolling friction as per a newton meter. Hence, the force outputted by the motor is sufficient for smoothly pulling the current model even with a 1:1 gearbox. However, to leave open the possibility for testing more mediums in the future, such as a full clay model or a snow model, the spooler can provide a maximum of 65 N pulling force with the 1: 6.25 gearbox if required, assuming an ideal gearbox. Even a 75% efficient gearbox would still be able to provide 48.75 N of pulling force.

The spooler is driven by an RS Hybrid Stepper Motor (1805279) and a DRV8825 motor driver board, with control logic and serial inputs received on an ESP32 microcontroller board. This allows control of speed and direction as well as revolving by single motor steps if needed, all depending on the serial output produced by the GUI upon a user's button press to start the collection of A-scans.

The SolidWorks assembly model of the winch is shown in Fig. 29, the winch was designed using the datasheet and a SolidWorks copy of the hybrid stepper motor [32].

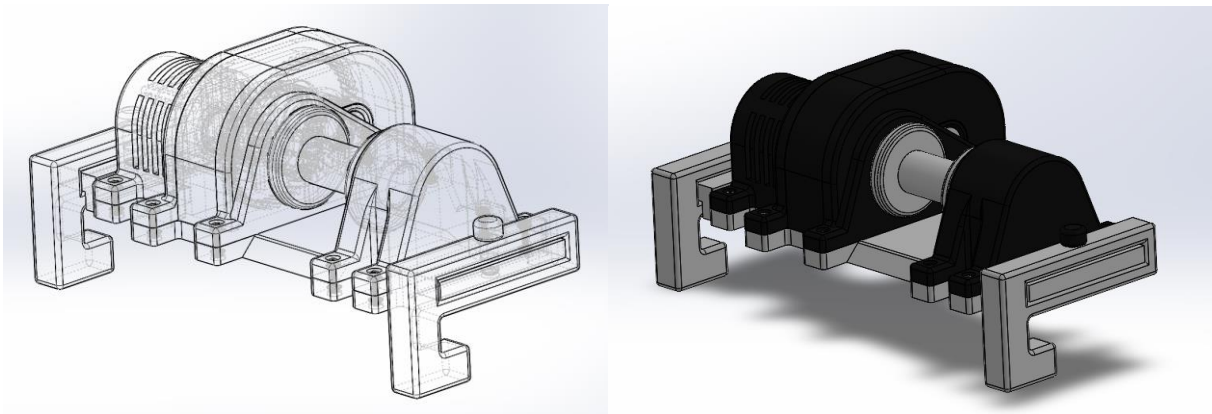


Fig 29. Spooler SolidWorks image (wireframe on the left and full image on the right).

The final model features four wheels on each corner, reducing the friction, allowing the weight to be disregarded when a high-torque driver motor is being used. Plastic rails were added to guide

the test medium antennas' centre, increasing the experiment's repeatability. The cross-section and isometric view is shown in Fig 30 and Fig 31.

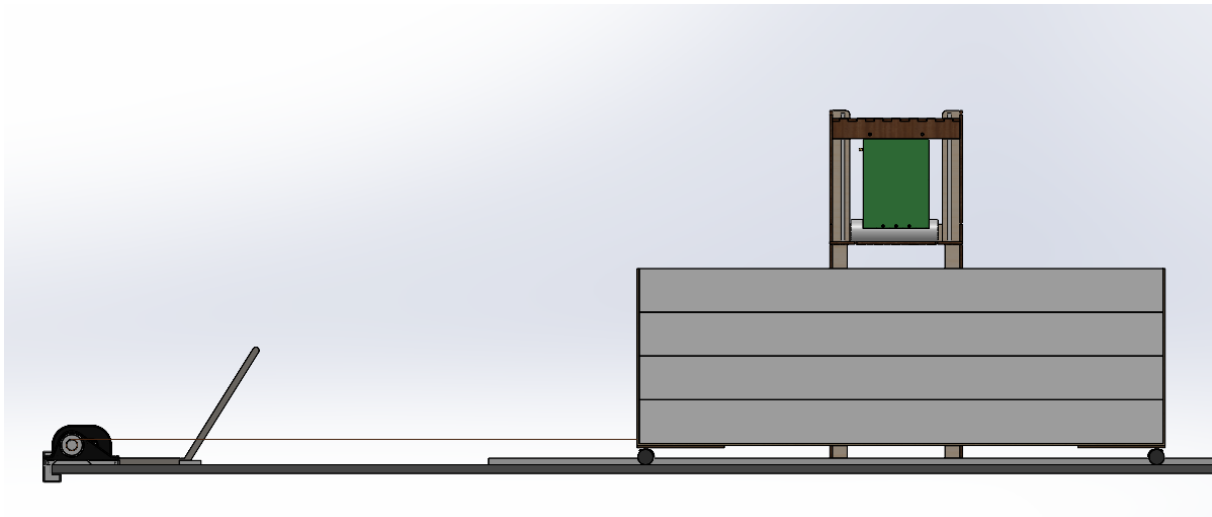


Fig 30. *Final test rig side cross-section view*

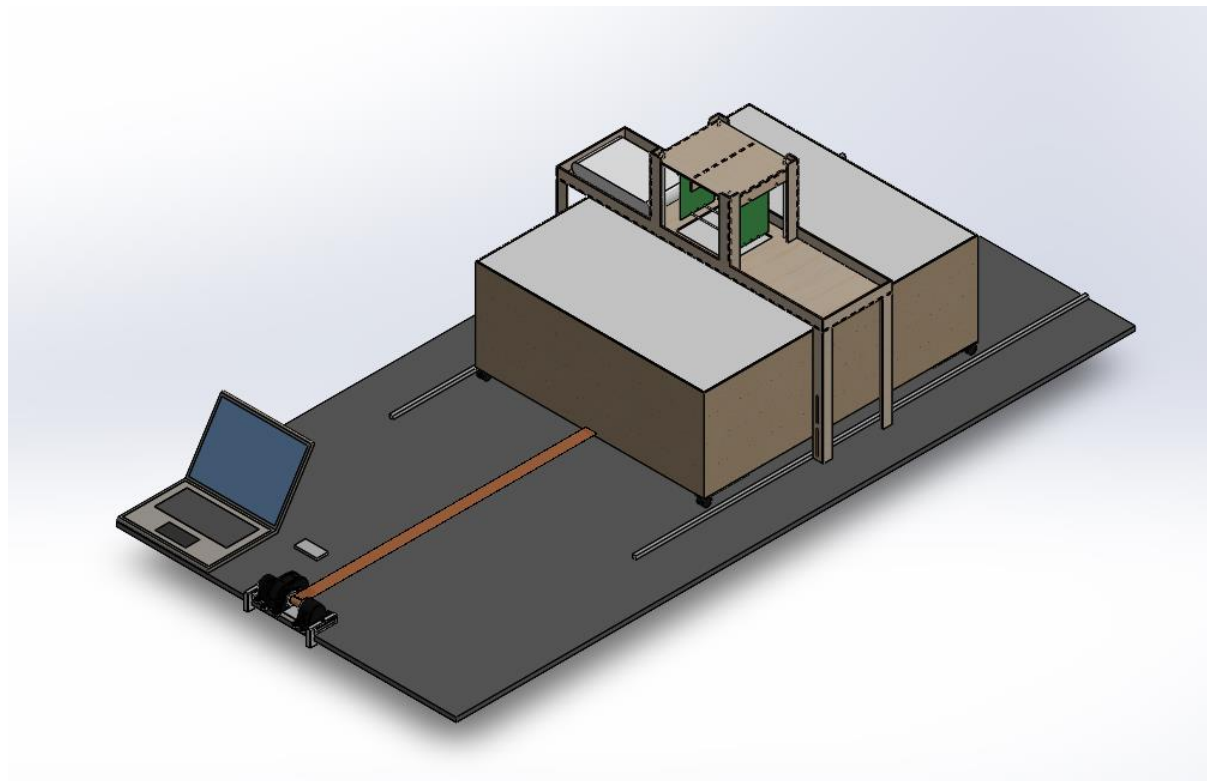


Fig 31. *Final test rig isometric view.*

2.3.2. Testing Medium

The englacial zone should typically be entirely made up of ice, although it can contain rock debris, water-filled cavities (particularly in temperate glaciers), and air-filled spaces. Other contaminants, which may be introduced through precipitation falling from the atmosphere or carried onto the snow surface in the accumulation area, also get integrated into the englacial region. Ice sheets, and to some extent polar glaciers, usually lack significant englacial inclusions, as shown by ice radar studies [33].

Research on the thickness of snow bridges over crevasses reveals considerable variability. For instance, TerraSAR-X satellite imagery captured a 6m thick snow bridge covering an 8m wide

crevasse in the McMurdo Shear Zone of Antarctica [34]. Additionally, field observations from the 2006 Issaqueena Ski Patrol exercise recorded a 6m span with a 2m thick bridge that showed slight sagging under its own weight [35]. Furthermore, a ground-penetrating radar survey of 912 crevasses at Totten Glacier indicated that snow bridges up to 4 meters thick were nearly always detectable, while some of the thickest bridges reached about 20 meters, though detection rates for these extremes were notably lower [36].

The medium of a crevasse mainly consists of snow at the surface and ice at the bottom, while the crevasse itself contains air and may have water at its base. However, during the experiment, the project used ice as a model and added a plastic box to simulate the crevasse. It was discovered that this approach caused the ice to melt at room temperature, increasing the amount of water around the model over time, which led to inconsistent experimental data. Additionally, the project did not have enough funds to conduct our experiments at temperatures below 0 degrees, so the group opted for a more straightforward and dielectrically similar approach.

To simulate subsurface discontinuities for crevasse detection, a layered testing medium was constructed using Styrofoam with a clay core insert. This setup was designed to mimic the dielectric contrast of an inverted crevasse when scanned with the GPR system.

Styrofoam was selected as the base material due to its low dielectric constant which is 1.1 [37], which approximates air which is 1.0 [38] in radar behaviour.

A Clay segment, centrally embedded in the Styrofoam, represented a high-contrast void or anomaly. Clay has a significantly higher dielectric constant which is 5.12 [39], similar to ice, creating a strong reflective boundary when encountered by the radar pulse.

The overall structure is relatively lightweight, repeatable, and suitable for repeated scans without deformation. The clay core was shaped to fit perfectly into the insertion gap to mimic an inverted cuboidal crevasse, ensuring a consistent radar signature. The structure can be seen in Fig 32.

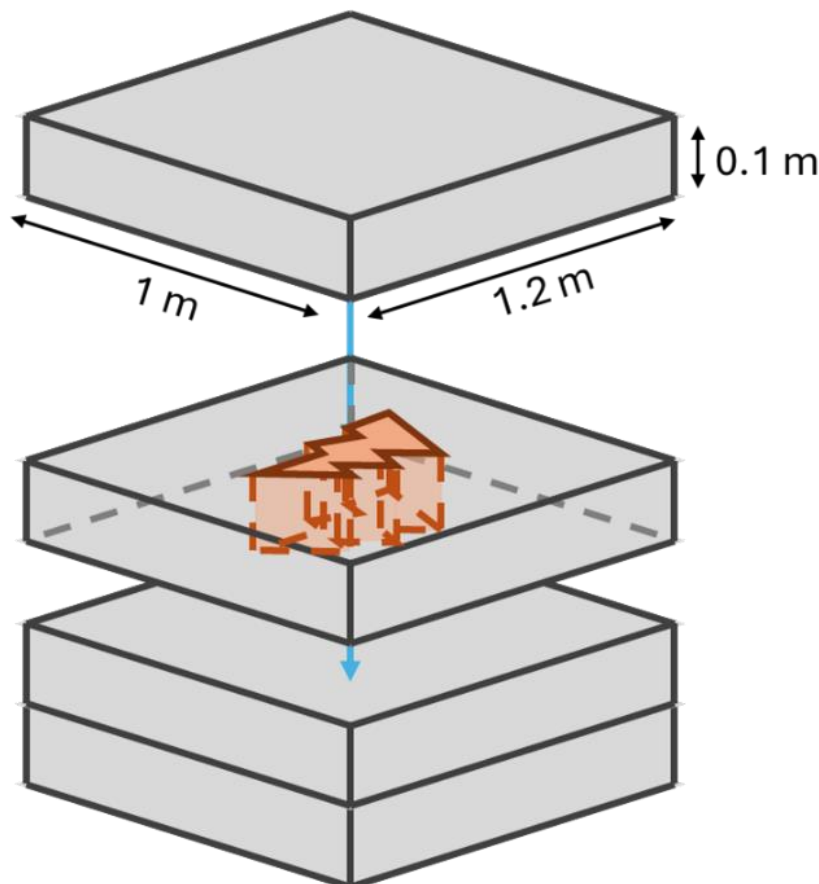


Fig 32. *Constructed Testing Medium Cross-Section.*

This controlled setup allowed for reliable generation of B-scans with identifiable crevasse-like features, which were then used for testing both signal consistency and object detection model accuracy.

2.3.3. UAV control simulation

The UAV control simulation module was developed to enable operator testing and demonstration of real-time flight control of a simulated UAV in a controlled environment. This builds upon the initial testing rig (Section 3.3.1), which was designed to simulate the relative motion of a GPR system for B-scan formation. While the early rig was focused on hardware-level signal acquisition and localisation, the UAV control simulation was introduced as a complementary software-level tool for interfacing with and controlling a virtual drone. This demonstrated that the “All in one” UI can control a simulated drone, and hence this can translate to the control of an actual drone in field use.

The control simulation uses the Ignition Gazebo simulator [40], a physics-based robotics environment, integrated with a custom Python-based user interface [41], [42]. The control interface allows manual and semi-autonomous flight of a Crazyflie UAV model, the Unified Robot Description Format (URDF) sourced from a GitHub repository [43], [44]. Users interact with the drone via common controllers (Xbox, PS5, etc.), with velocity commands mapped to joystick axes and transmitted to the UAV model in real time.

A checkbox-based toggle is provided for simulating with or without physics. In non-physics mode, a custom Ignition Gazebo plugin disables dynamics, allowing visual demonstrations without interference from physical instability. Logs, controller feedback, and telemetry outputs are displayed on the GUI to aid debugging and user feedback, as shown in Fig 33.

Key features of the UAV control simulation include:

- Manual control via joysticks.
- Start/Stop buttons are used to initiate and terminate the UAV simulation environment.
- Command logging with real-time visual feedback.
- Integration with the GPR system for radargram updates during flight.

This simulation tool serves as a standalone demonstration of the UI's capability and as an intermediate testing layer before real-world UAV deployment.



Fig 33. GUI demonstrating successful drone control.

2.3.4. UAV Design

Achieving airborne GPR with UAVs is tough in extremely cold and windy conditions, due to concerns regarding aerodynamics and electronics. Therefore, some requirements for temperature control and propeller protection are needed. This section further examines the UAV design for polar environments and investigates the best solution.

EastGRIP Camp

For a research team in a polar environment. During their time at the EastGRIP camp in 2018, the team executed a proof of concept for airborne sampling and surface analysis by utilising a compact remote-controlled sampling unit and a multi-rotor UAV (DJI S-1000, DJI, Inc.) [45]. The system successfully gathered data and samples for analysis up to 400 m, but it faced significant challenges with navigation and control due to its closeness to the magnetic pole and battery performance in low temperatures, which restricted flight durations to under 15 minutes [45].

The S2 fixed-wing aircraft was selected for the 2019 campaign. Black Swift Technologies, LLC designed the S2 is a modular, autonomous, electric-powered aircraft with a lightweight composite airframe [45]. Thanks to its pneumatic launch system, it can perform fully autonomous flights in rugged areas and adjusts to changing wind conditions for stability during mapping or atmospheric sampling [45]. The aircraft carries up to a 3.5 kg payload for up to 90 minutes, though in arctic conditions, typical flight time was 45 minutes, sufficient for climbing 1600 m and conducting necessary sampling [45].

ALICE

The quadcopter ALICE (Airborne tool for methane isotopic composition and polar meteorological experiments) was designed to measure temperature, humidity, solar irradiance, and surface temperature, as well as to collect air samples for lab analysis [46]. It features a central plate and four detachable arms with T-Motor U11 120 KV motors and T-Motor 26 × 8.5 CF propeller blades, allowing for easy assembly. The central plate houses LiPo batteries, the PixHawk autopilot, and sensors. Flights were conducted in line of sight without de-icing systems [46]. All components were tested in a climate chamber at -40°C, and vibration tests were performed [46]. ALICE was utilized during the Polarstern expedition (ARK-XXXI/4) in the northeast Greenland Sea from 16 September to 14 October 2017, completing eight flights up to 1000 m with a total flight time of 1 hour and 7 minutes [46].

ALADINA

The ALADINA (Application of Light-Weight Aircraft for Detecting in-situ Aerosol) system is based on the “Carolo” unmanned vehicle, designed to carry additional payload for measuring boundary layer aerosol particles [46]. It features a 3.6 m wingspan, a maximum take-off weight of 25 kg (including a 4.5 kg payload), and a pusher configuration to minimize flow disturbance at the measurement point [46]. Powered by a 3.8 kW electric motor for agile flying, it operates without polluting emissions. The aircraft has a flight duration of up to 40 minutes, with a battery capacity of 1 kWh, cruising at 28 m/s in winds below 15 m/s. All avionics are powered by redundant systems for safety, using the Pixhawk 2.1 Cube autopilot [46]. During a campaign in Svalbard from April 24 to May 25, 2018, ALADINA completed 50 flights totalling 30 hours and collected 220 vertical profiles up to 850 m ASL [46].

Comparison Between Fixed Wing & Quadcopter UAV

Fixed-wing models excel over rotary-wing drones in flight endurance, design simplicity, and operational costs [47]. Due to their longer endurance and faster speeds, fixed-wing models are ideal for extensive area surveys, making them preferable for meteorology, atmospheric science, and glaciology applications [47]. In contrast, rotary-wing drones offer better payload capacity and the ability to hover, which is essential for photography [47]. Their lower speeds and greater manoeuvrability make them suitable for challenging terrains, and they are easier to operate as they don't require specialised launch equipment [47].

Table 3. *Pros and cons of fixed & rotary wing UAV.*

Comparison Aspect	Fixed-Wing Pros	Fixed-Wing Cons	Rotary-Wing Pros	Rotary-Wing Cons
Endurance	Long range (~90 min; ~45 min cold)	No hover; constant forward speed	Hover capability	Short range (≤ 15 min)
Speed	Fast cruise (~28 m/s)	Poor for fine sampling	Precise low-speed flight	Slow coverage
Manoeuvrability	Stable straight flight	Wide turn radius	Agile, obstacle avoidance	Sensitive to gusts
Take-off & Recovery	Launcher or runway	Needs open space & de-icing	Take-off from any site	Requires rotor de-icing
Structure & Maintenance	Simple airframe, low upkeep	Composite repairs need specialists	Modular parts, easy swaps	Many moving parts
Payload	Up to 3.5 kg	Mounting limited by wings	Flexible mounts	Max ~2 kg
Polar Suitability	Fuselage can be insulated	Battery loss; magnetic interference	Redundant batteries; cold-tested	Rotor icing; limited in high wind

After comparison from Table 3, while fixed-wing platforms such as the S2 and ALADINA offer superior endurance and faster cruise speeds—advantages for large-scale surveys and long transects—their reliance on pneumatic launch systems, complex recovery procedures, and sensitivity to magnetic disturbances in polar regions limit their practicality. In contrast, rotary-wing quadcopters like ALICE deliver the agility to hover precisely for targeted sampling, carry moderate payloads, and operate from confined or uneven terrain without specialised launch gear. Their simpler take-off and landing requirements, combined with proven performance in sub-zero temperature chambers and robust battery management, make quadcopters the more adaptable choice for our polar GPR application. Consequently, our team will adopt a quadcopter configuration to ensure reliable, repeatable airborne surveys and sample collection under the extreme cold, wind, and logistical constraints of polar fieldwork.

2.4. Machine Learning and Crevasse Detection

The project integrates a machine learning pipeline to enable automatic identification of crevasses in Ground Penetrating Radar (GPR) radargrams. This capability supports fast and objective post-processing of radar data, reducing the reliance on manual interpretation. The detection system is implemented using a YOLO (You Only Look Once) [48] object detection model trained on a custom dataset of GPR images. Fig 34 shows a section of training and validation set in YOLO crevasse detection model.

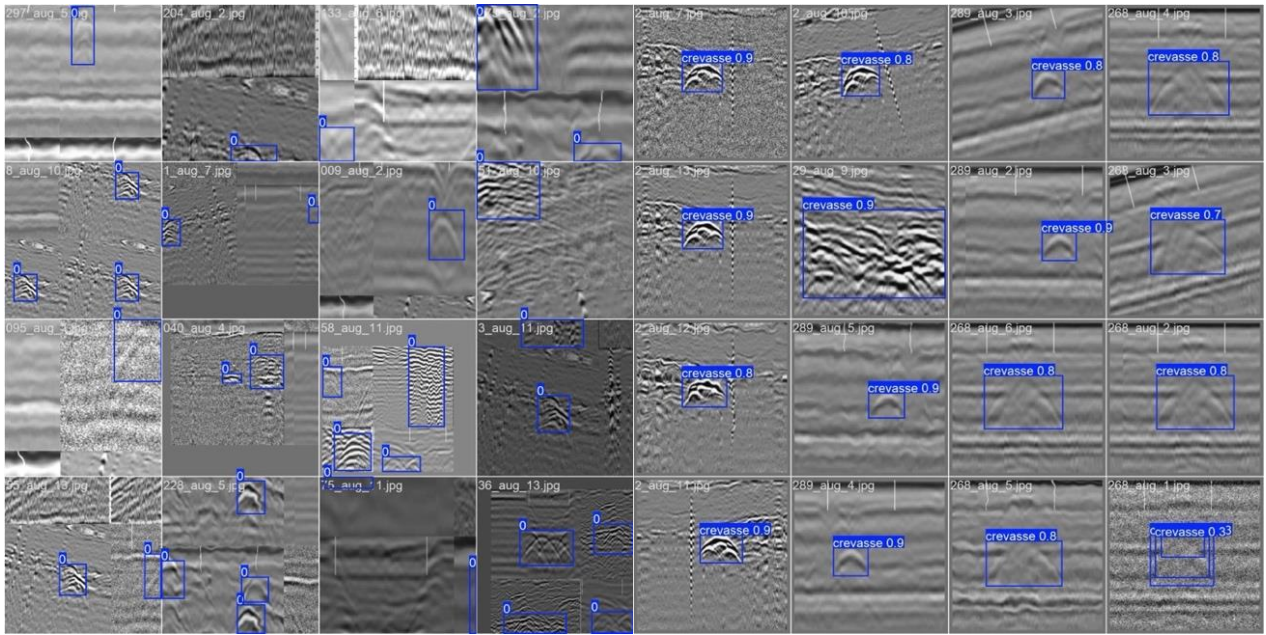


Fig 34. Image of train (left) and validation (right)

2.4.1. YOLO

YOLOv8 demonstrates impressive real-time detection capabilities, processing images at roughly 97 frames per second (FPS), which significantly surpasses the performance of Faster R-CNN and SSD, which are 8 FPS and 60 FPS respectively [49]. Additionally, with an inference time of approximately 10.3 milliseconds per image [49], YOLOv8 is ideally suited for applications that require immediate responses, such as autonomous driving and surveillance systems.

Competitive accuracy shows that while Faster R-CNN achieves a slightly higher mean Average Precision (mAP) of 89.3%, YOLOv8 closely follows with an mAP of 85.6% [49], offering a balanced trade-off between speed and accuracy. Additionally, in tasks that require precise object localization, YOLOv8 demonstrates strong regression capabilities, making it highly effective for applications such as object tracking and pose estimation [49].

Additionally, in future projects it may be necessary to be able to detect crevasse in real-time environments where YOLO can monitor the B-scan in real-time. The full comparison is in Table 4.

Table 4. Comparison between YOLO, R-CNN, and SSD

Model	mAP (%)	Inference Time (ms)	FPS	Notable Strengths
YOLOv8	85.6	10.3	97	Real-time detection, versatile applications
Faster R-CNN	89.3	120.4	8	High accuracy, detailed object localization
SSD	78.2	16.5	60	Balanced speed and accuracy

2.4.2. Data Preparation

A custom dataset was compiled from radargram images representing various subsurface features, including underground cavities [50]. The dataset includes both raw and augmented images to improve model generalization. Images were organized into training and validation sets, with the following characteristics:

- **Training Set:** Contained radargram images labelled with bounding boxes centred on known crevasse locations. Data augmentation techniques (e.g. flipping, noise injection) were applied to increase sample diversity.
- **Validation Set:** Used to evaluate model generalisation during training.
- **Class Labels:** Due to the specific nature of the detection task, the model was trained on a single class, “crevasse.”

Bounding boxes were annotated appropriately, ensuring the bounding box encapsulated only the target feature.

2.4.3. Model Selection and Training

The Ultralytics YOLOv8 [51] architecture was selected due to its balance of inference speed and accuracy, and is suitable for real-time applications.

The test found that although YOLO has powerful processing performance, it is still challenging for the CPU of high-performance computers for home use. Typically, an iteration takes 5 to 6 minutes, drastically limiting the number of epochs. Furthermore, the CPU is less rapid or effective than the GPU for image processing. Therefore, there is a method to utilise NVIDIA CUDA [52] to train the model using GPUs. This approach significantly enhances training efficiency, reducing the time to 25 to 30 seconds per epoch. This approach is provided by [53], and the comparison between CPU and GPU is shown in Table 5. [53].

Table 5. *Comparison between CPU and GPU in YOLO Model [53]*

Aspect	CPU	GPU
Inference Time	~500 ms/image	~15 ms/image
Training Speed	~2 epochs/hour	~30 epochs/hour
Batch Size Capability	Small (2–4 images)	Large (16–32 images)
Real-Time Performance	No	Yes
Parallel Processing	Limited	Excellent (thousands of cores)

- **Model Architecture:** YOLOv8s (small variant) pre-trained on COCO dataset.
- **Training Configuration:**
 - Epochs: 100
 - Batch size: 1339
 - Input image size: 640 × 640
 - Optimiser: auto [54]
 - Learning Rate: 0.01 [54]
- **Hardware:** Training was conducted on a local GPU-enabled workstation.

The trained model achieved high confidence in detecting crevasse features within the radargram dataset, with performance evaluated using mean Average Precision (mAP) and recall.

2.4.4. Detection Interface Implementation

A graphical interface was developed using TKinter [55] to allow users to apply the trained model to radargram images. The interface supports the following workflow:

In this process, the user selects a radargram image from the local directory, which the YOLO model then processes to identify crevasse features. The model’s inference results in bounding boxes overlaid on the radargram, visually indicating the detected features. Finally, users can export the annotated image for documentation or further analysis, enhancing their ability to study the crevasse detections comprehensively. The whole GUI tab pages are shown in Section 2.5.

2.4.5. Detection Accuracy and Limitations

Confusion Matrix

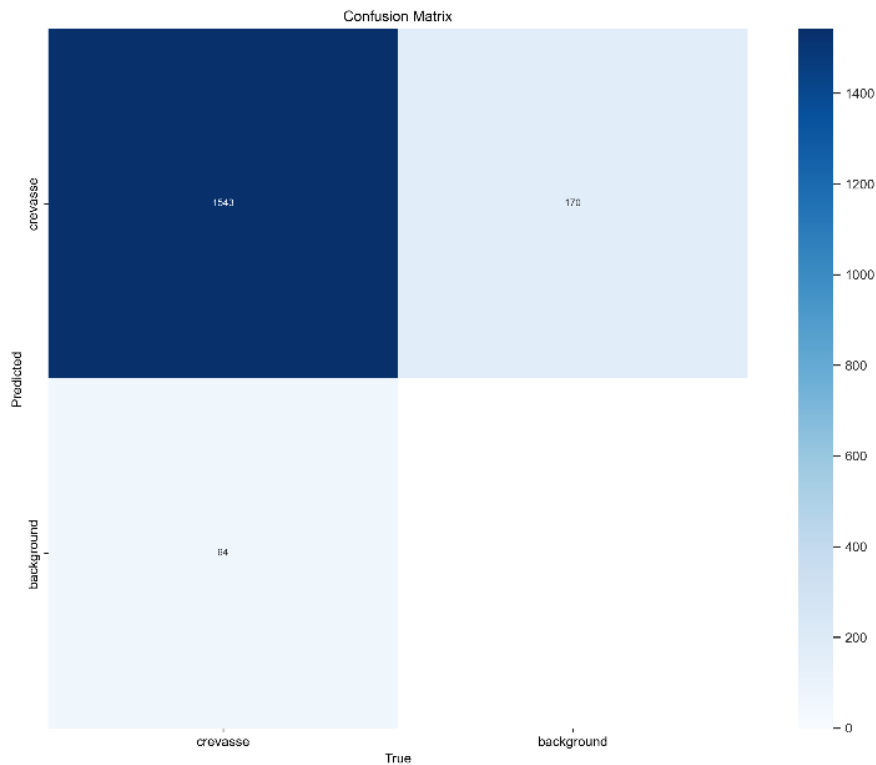


Fig 35. Confusion matrix of YOLO crevasse detection model

In the raw confusion matrix shown in Fig 35, the “crevasse” class demonstrates a strong performance with 1543 true positives and only 170 false negatives, alongside 64 false positives. The normalised matrix further highlights this success, with “crevasse” achieving approximately 96% recall, while the background class approaches nearly 100%, showing very low misclassification rates. This is significant as the high number of true positives, coupled with low false positives and negatives, indicates that the model is effectively localising and classifying objects, while maintaining controlled rates of missed detections and false alarms.

Precision-Recall Curve

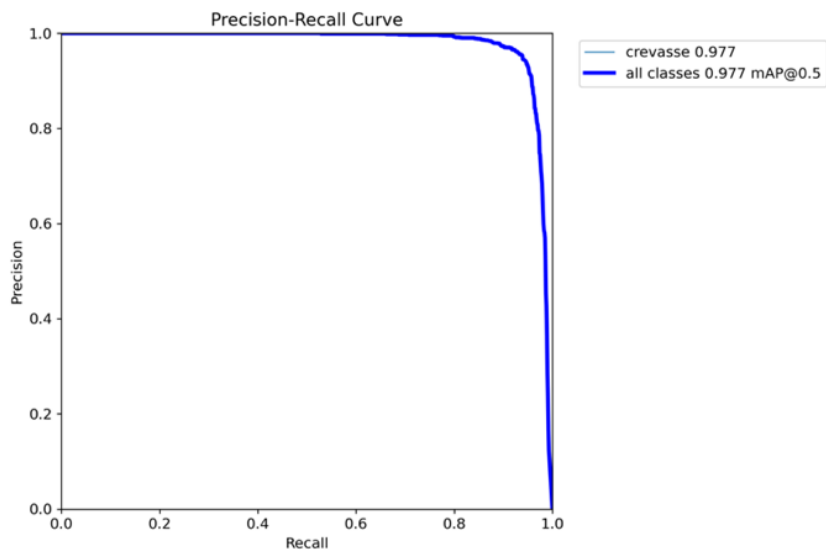


Fig 36. Precision-Recall curve of YOLO crevasse detection model

From Fig 36, the area under the PR curve is 0.977. This performance is demonstrated by the curve, which hugs the top-right corner, indicating that precision remains consistently high across a broad range of recall values. This is significant because the mean Average Precision (mAP) serves as the standard summary metric for object detection; a high mAP indicates that the model is proficient in both detecting targets (recall) and accurately localising and classifying them (precision).

Results

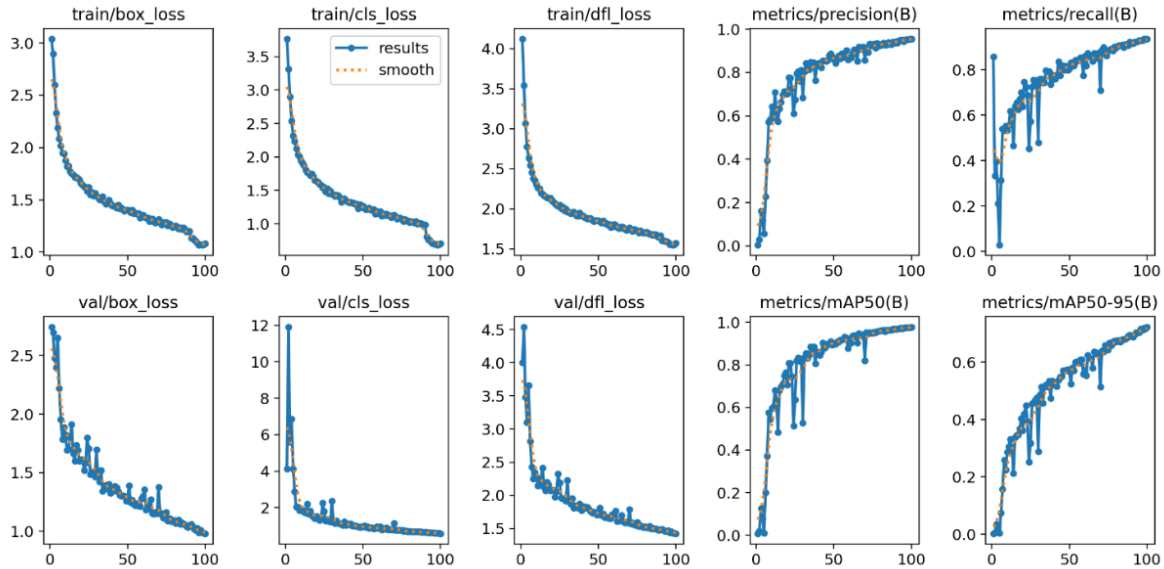


Fig 37. Results diagrams of YOLO crevasse detection model

From Fig 37, the box, classification, and DFL losses decrease smoothly and in sync on both training and validation sets, showing no large oscillations or divergences, which indicates good convergence and minimal overfitting. Additionally, precision, recall, and mAP curves rise steadily and stabilise at high values. This is significant because these learning curves confirm that the model genuinely learns useful features, as evidenced by decreasing loss and increasing metrics, rather than merely memorising noise, which suggests that it generalises well to unseen data. Fig 35, Fig 36, and Fig 37 are produced by the YOLO training model itself.

Summary

In summary, these plots together demonstrate that the YOLO model attains high true-positive rates, low false-positive and negative rates, exceptional overall detection accuracy (high mAP), a strong balance between precision and recall, adaptable thresholding, and smooth, stable training dynamics – compelling evidence that this is a high-quality model.

2.5. Graphical User Interface

The graphical user interface (GUI) is the primary access point for operating the UAV-mounted GPR system and analysing radargrams. It encapsulates system control, real-time visualisation, and machine learning-based crevasse detection into one application. The GUI is split into modules like a UAV and GPR control interface built using, and a radargram analysis interface using with integrated YOLO object detection. The GUI is coded in Python.

2.5.1. Data Collection Tab

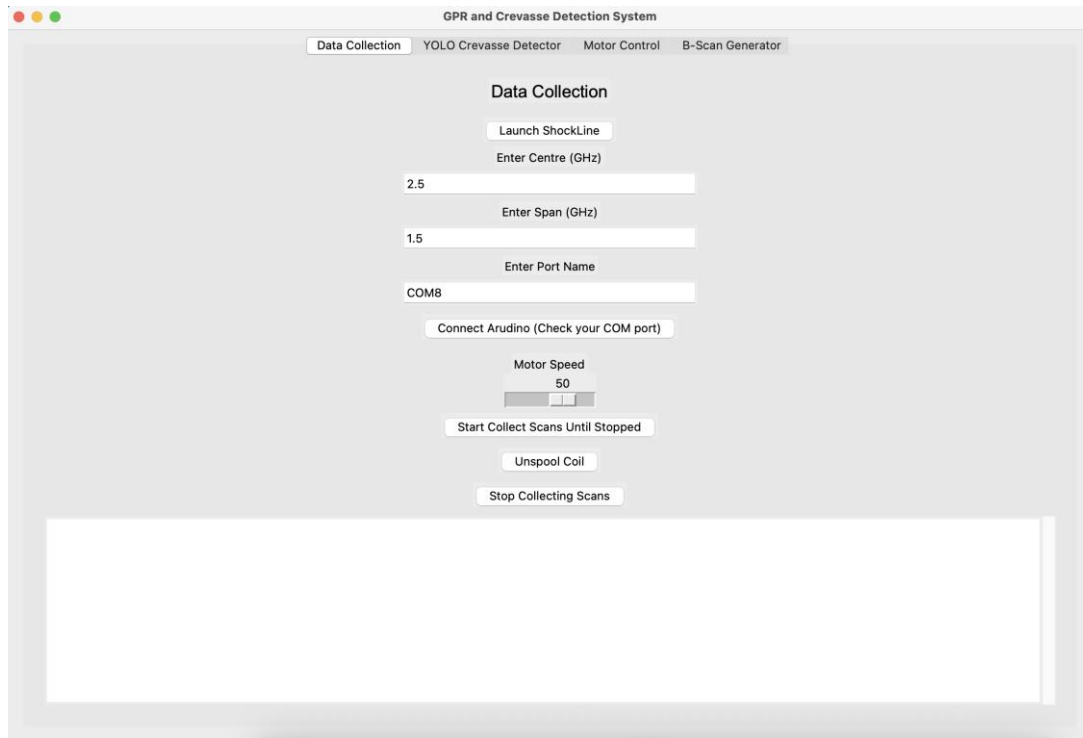


Fig 38. *Data Collection tab view*

The data collection tab serves as the project's main interface, gathering data of A-scans. The overall arrangement is structured from top to bottom, allowing users to complete the entire operation simply by clicking a button or entering data step by step. Fig 38 illustrates the view of the “Data Collection” tab.

The “Launch ShockLine” button can automatically open the Anritsu VNA’s ShockLine software on a Windows PC. The user needs to set the centre and span of the antenna frequency. Next, the user must enter the port name of their Arduino. After that, click to connect to the Arduino. The Arduino needs to connect to the PC. If there is no error, an error will be shown in the log. The user can adjust the speed of the motor. Once this is done, the user can click “Start Collecting Scans”; the motor will automatically begin collecting scans of the object. After the scan, the user can unspool the coil to initiate the object. Finally, the stop button can halt the motor. A log scroll view is below all of the buttons; the feedback or error of each button is shown in the log area.

2.5.2. Optional Tabs (Motor Control & YOLO Crevasse Detection Tab)

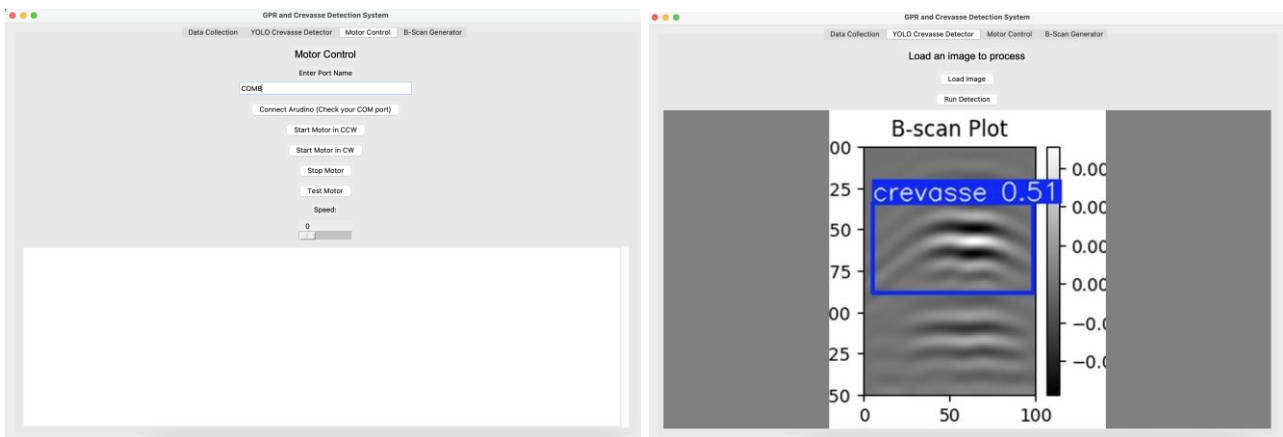


Fig 39. *Motor Control(left) & YOLO crevasse detection(right) tab view*

As Fig 39 left shows, the Motor Control tab is not a mandatory page for the user. It contains all the operations on the motor and can be used by the user to test the motor to ensure it is working correctly. CCW means that the motor rotates anticlockwise, and CW means that the motor rotates clockwise. All other buttons have the same effect as their names.

The YOLO crevasse detection tab allows the user to select any image for deep learning crevasse detection. The image will automatically generate output.png in the same folder. The Fig 39 right shows the YOLO crevasse detector tab and also highlights its successful functioning on a B-scan generated by the SSPOC. This tab is intended for importing existing plots, machine learning can already be applied to freshly generated plots in the B-scan generator tab.

2.5.3. B-scan Generator Tab

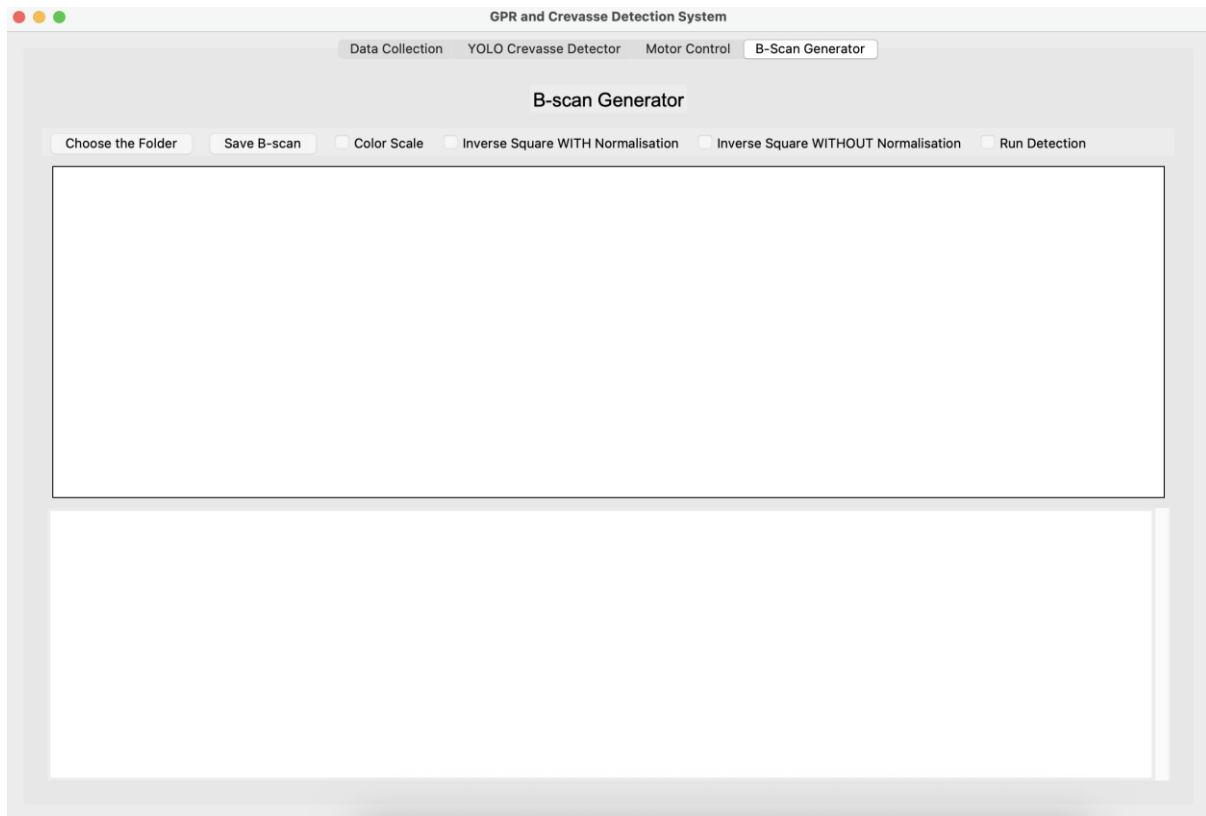


Fig 40. *B-scan Generator tab view*

The B-scan generator tab allows users to convert A-scan folders into B-scans directly. Firstly, click the "Select Folder" button in the top left corner, then choose the parent folder of the A-scan folder. This action will display all A-scan folders within that folder in the interface. By double-clicking on the folder name, the B-scan will be automatically generated and shown at the bottom of the interface. At this stage, when the user clicks any checkbox, the displayed B-scan will change according to the user's selection. All checkbox functions correspond directly to their labels. After all, users can click the save B-scan button to save the B-scan to any of their files. Thus, a user can select from several different display types of the B-scan data. The interface can be seen in Fig 40, and the output image can be seen in Fig 41.

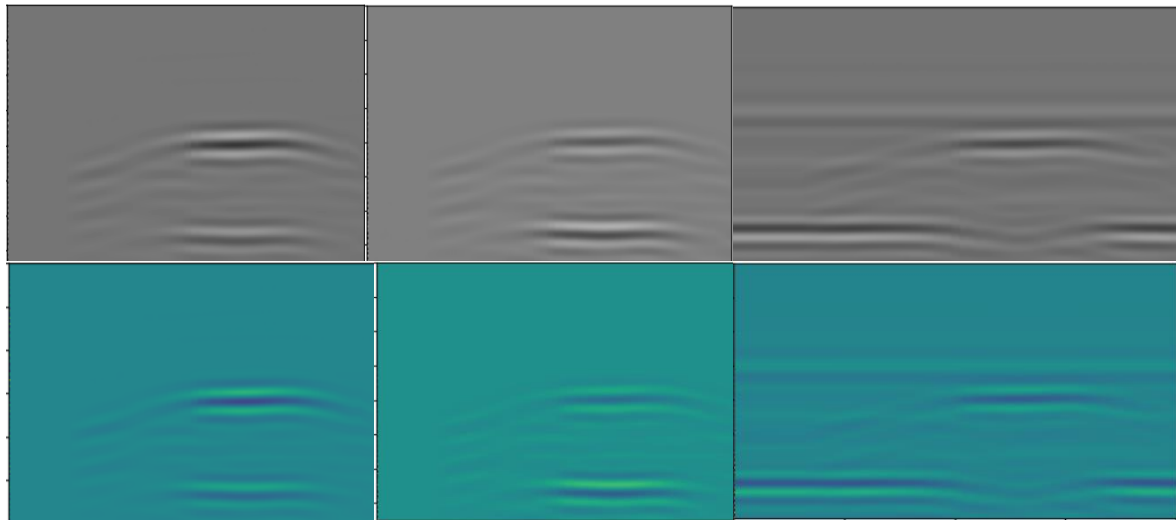


Fig 41. Example of same B-scan output with different signal processing/ heatmaps applied.

2.5.4. Implementations And Technologies Used

The GUI combines multiple programming libraries and systems for modular operation, as shown in Table 6. For all library providers: [41], [42], [55], [56], [57], [51], [58], [59] and [60].

Table 6. Programming libraries

Feature	Implementation
Control Panel	PyQt5 & Tkinter
Real-time GPR Visualization	Matplotlib QtSagg
DualSense Controller Input	Pygame
Machine Learning (YOLO)	Ultralytics YOLO
Data Handling	NumPy
Communication Protocol	SciPy
Serial Port	pySerial

The architecture is modular and designed to support future upgrades such as live model inference, field hardware integration, and autonomous path planning for the UAV.

2.5.5. System Interaction and Workflow

The intended user workflow is as follows in Fig 42. It lay out the steps a user would take to run a demonstration of the SSPOC. This workflow can translate to a field use of the GUI and GPR hardware if we account for the drone control being modified to control a real drone, the localisation method being substituted for one onboard a UAV, and the serial interfaces being changed from wired to wireless ones.

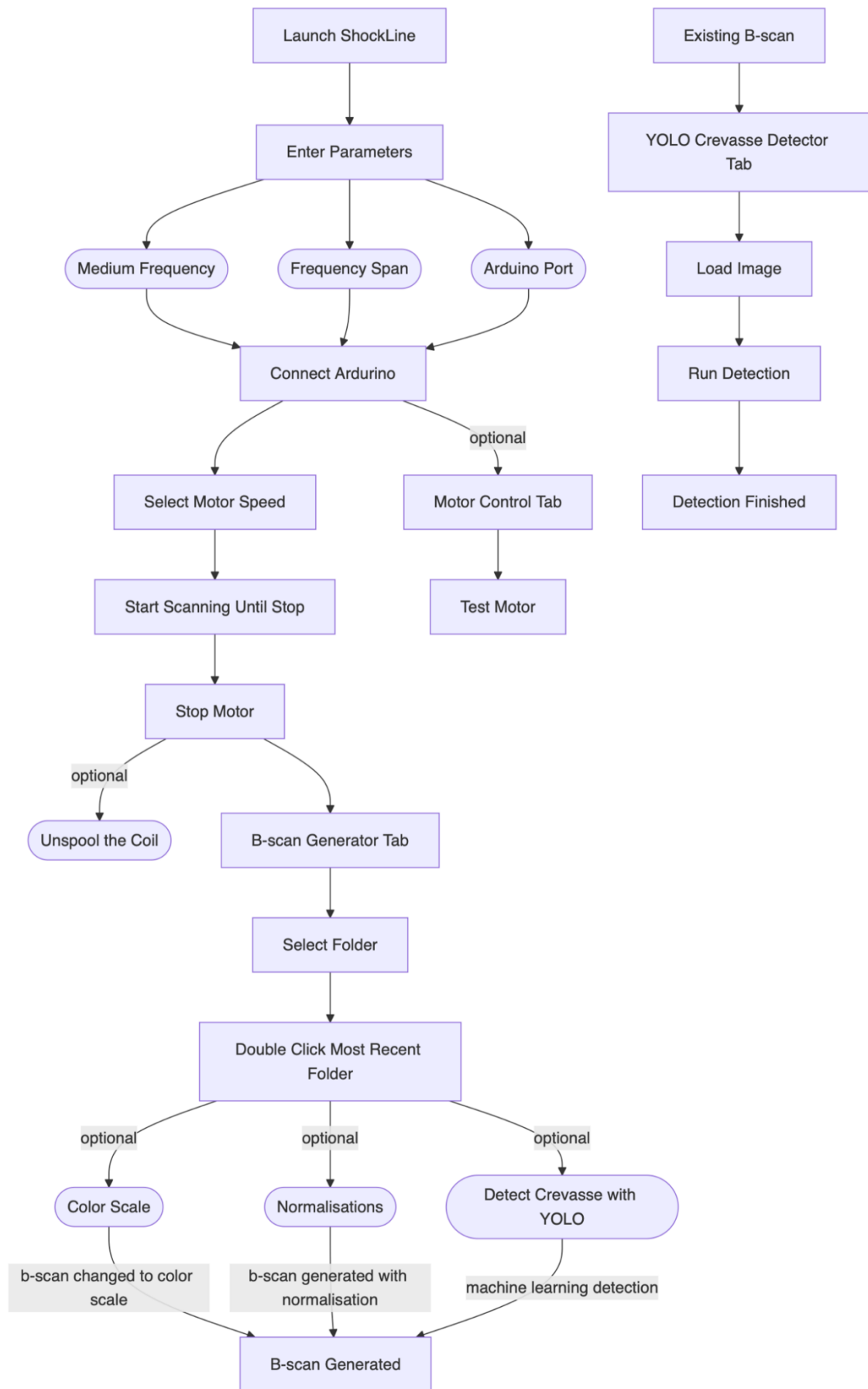


Fig 42. Typical User Workflow

2.5.6. Challenges and Considerations

The challenges and considerations for crevasse detection revolve around a few key aspects: currently, live inference is not implemented, as detection is performed on static radargrams, with real-time detection during flight suggested as a future enhancement. Additionally, some joystick axes require calibration to prevent unintentional movements during simulations. While YOLO detection is fast, it does not support real-time processing for large datasets or batch operations, indicating the need for further optimisations for deployment on low-power embedded systems. Furthermore, the system relies on a controller, an appropriate simulation environment, and specific Python dependencies, which may hinder accessibility and ease of use.

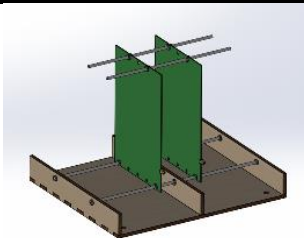
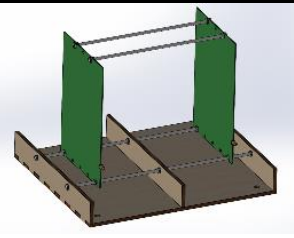
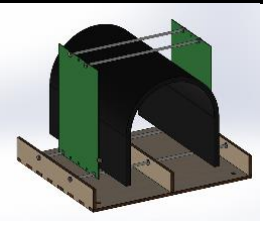
2.6. Testing and Results

Testing of the SSPOC as well as the Bespoke GPR prototype was carried out to validate their functionality, accuracy, and robustness. Tests for the SSPOC involved antenna selection and antennae separation selection including range testing, verifying the effectiveness of the signal processing pipeline in generating interpretable radargrams, and evaluating the accuracy of the YOLO machine learning model. The bespoke GPR prototype tests involved assessing the prototype's signal transmission and reception capabilities and hence the resulting A-scans. Results from these tests informed iterative design improvements and demonstrated the overall feasibility of the system.

2.6.1. Testing the SSPOC

The group first carried out a clutter analysis, like what was done in Fig 47, to deduce the cross-coupling signal seen at when the Vivaldi antennas were placed at a separation of 10 cm and 20 cm. A smaller cross coupling signal was desired by the group, as this would mean that the ranges in which cross-coupling could interfere with detected targets would be smaller. As seen in Table 7 the larger separation of 20 cm showed a smaller cross coupling signal, and this was further reduced when radar absorption material was placed between the antennas.

Table 7. *Cross-coupling values for different antenna set ups*

Antenna Design	Vivaldi (10 cm separation)	Vivaldi (20 cm separation)	Vivaldi (20 cm separation with radar absorption material)
Picture Of Antenna Design			
Cross-coupling Range (cm)	25.5	24.0	22.5

Testing Anritsu GPR System

The group evaluated four CW frequency sweep bandwidths – including 20 MHz to 1 GHz, 2 GHz, 3 GHz and 4 GHz— on the SSPOC with both the Vivaldi antennas mounted 0.20 m apart and the V14 antennas. These spans showed theoretical promise in terms of their range resolution and

depth penetration, while also ensuring that the reactive region of each antenna would be unlikely to impede with the frequency sweeps. These theoretical results are summarised in Table 8.

Table 8. *Evaluation of different bandwidth*

Bandwidth (GHz)	Rang Resolution (cm)	Depth penetration in foam (m)	Depth penetration in ice (m)	Reactive (m)	Far-field (m)
0.02 – 1	15	435	120	< 0.047	> 0.077
0.02 – 2	7.4	217	61	< 0.066	> 0.152
0.02 – 3	4.9	145	41	< 0.081	> 0.227
0.02 – 4	3.7	109	31	< 0.093	> 0.302

As the 20 MHz – 1 GHz bandwidth did not have a depth penetration small enough to differentiate between the different layers, only the remaining bandwidths were tested with a several sized foils at 0.5 m and 1 m away from the antennas to gain an understanding of the behaviour of each of the antennas. S21 data was displayed in the time domain in 2 different manners, one displaying only the real data, while the other converting the data into log magnitude format. These tests would be necessary to help the group with key design decisions in not only testing the physical capabilities of the SSPOC on the test rig but also deduce theoretical behaviour of these waves in when passing through mediums that are common to polar regions (such as ice, snow, etc.). Each test kept a 1 m³ area around the antennas free of any mediums – not including Styrofoam and air – that could interfere with the gathered data of the S21 parameters, as shown in Fig. 43.

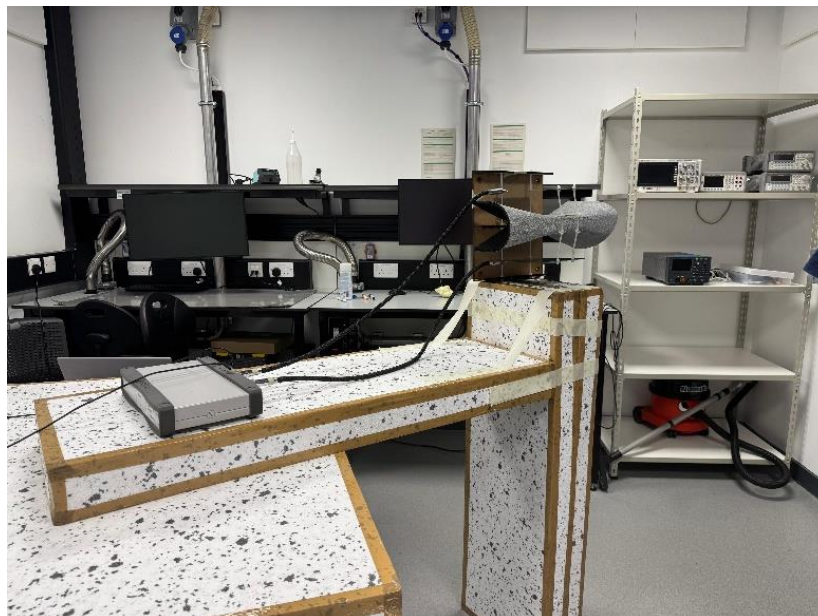
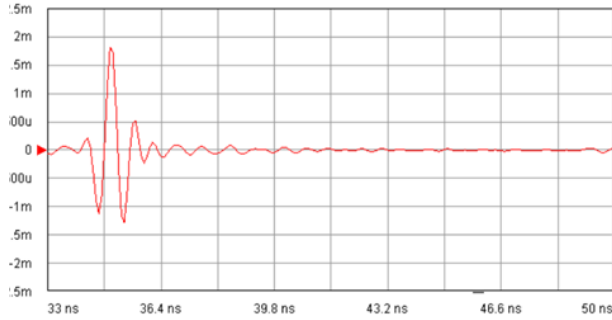


Fig 43. *Image of the test set up for initial tests, with 1-meter free space around antennas.*

The results of the experiments when using a 30 cm x 30 cm foil sheet, held 1 m away from the antenna, are shown in Fig 44. These initial tests allowed the group to conclude that the 20 MHz – 3 GHz and 4 GHz bandwidth frequency sweeps were better optimised in detecting targets at the narrow range needed for the test medium. Ultimately, the 20 MHz – 3 GHz bandwidth was chosen due to the reduced noise seen in smaller bandwidths [61]. It is also worth noting that the outputs of the S21 real data, were less noisy than the outputs from the Vivaldi's. This distinction between the reflection of the foil and background noise also suggested that the outputs of the V14 antennas would be less susceptible to errors from noise.

I S21 Trans Real RefLvl: 0 U Res: 500 uV/Div TBP



I S21 Trans Real RefLvl: 0 U Res: 10 mV/Div TBP

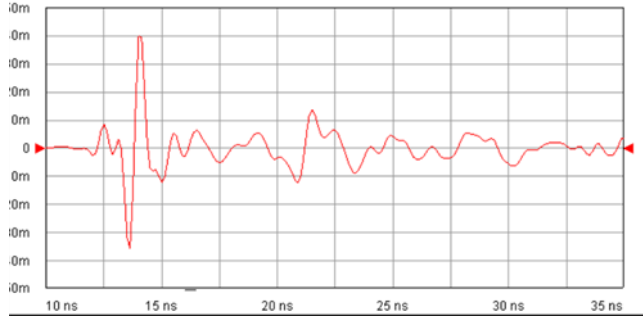
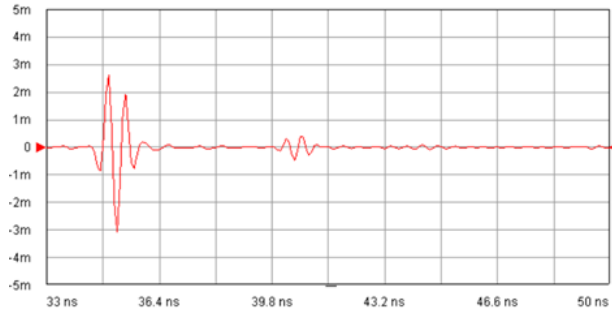


Fig 44. 20MHz-2GHz real S21 data, V14 Antennae (left) and Vivaldi (right)

I S21 Trans Real RefLvl: 0 U Res: 1 mV/Div TBP



I73 S21 Trans Real RefLvl: 0 U Res: 2 mV/Div TBP

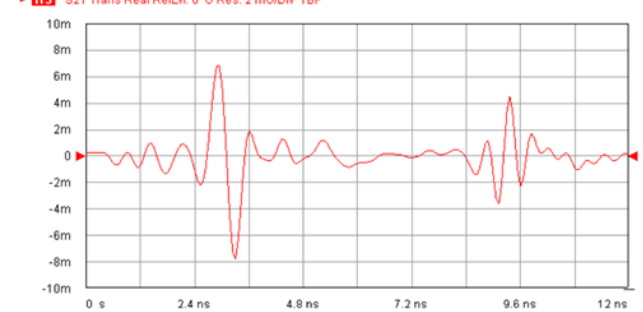
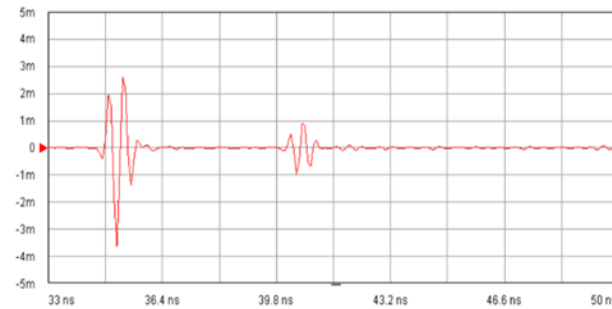


Fig 45. 20MHz-3GHz real S21 data, V14 Antennae (left) and Vivaldi (right)

I73 S21 Trans Real RefLvl: 0 U Res: 1 mV/Div TBP



I73 S21 Trans Real RefLvl: 0 U Res: 2 mV/Div TBP

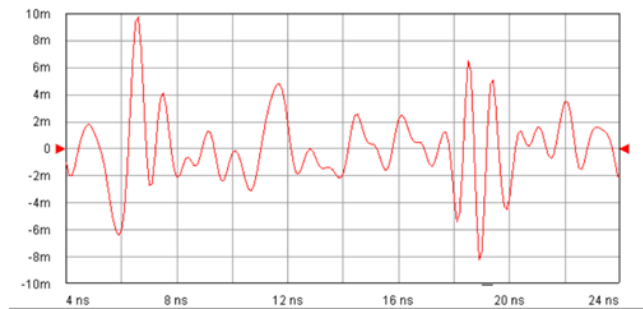


Fig 46. 20 MHz-4GHz real S21 data, V14 Antennae (left) and Vivaldi (right)

The clutter analysis of these waveforms, as shown in Fig 47, gave the group a clear definition of the size of the cross-coupling signals, their respective widths and the clutter signal for each antenna and bandwidth. The analysis of the clutter signals allowed the group to assume that signals above -90 dB and -50 dB could be detected by the V14 antennas and Vivaldi respectively. This would prove to be useful in detecting the theoretically deepest penetration these antennas could have when detecting crevasses in a real-world situation, when they are above snow or ice in the arctic (see Section 2.9).

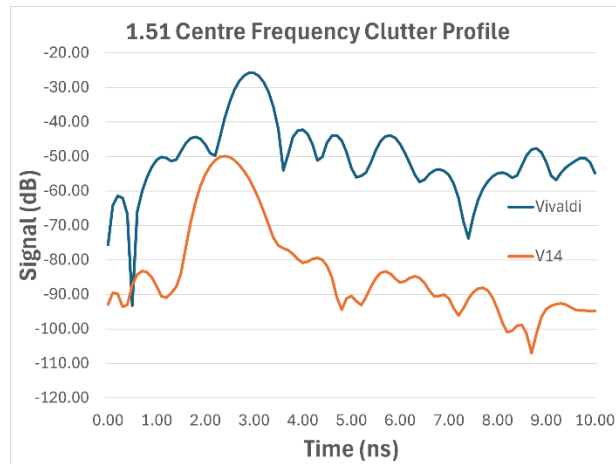


Fig 47. Clutter Profile of both antennas at the tested bandwidths

As the model contains 4 sheets of Styrofoam with a depth of 10 cm, foil was placed on top of the model, followed by placing it in between each layer of foam within the model. Successful tests would show a clear depth difference between each layer where the foil was placed. Fig 48 showed that both antennas were successfully able to differentiate between the different layers in which the foil was placed.

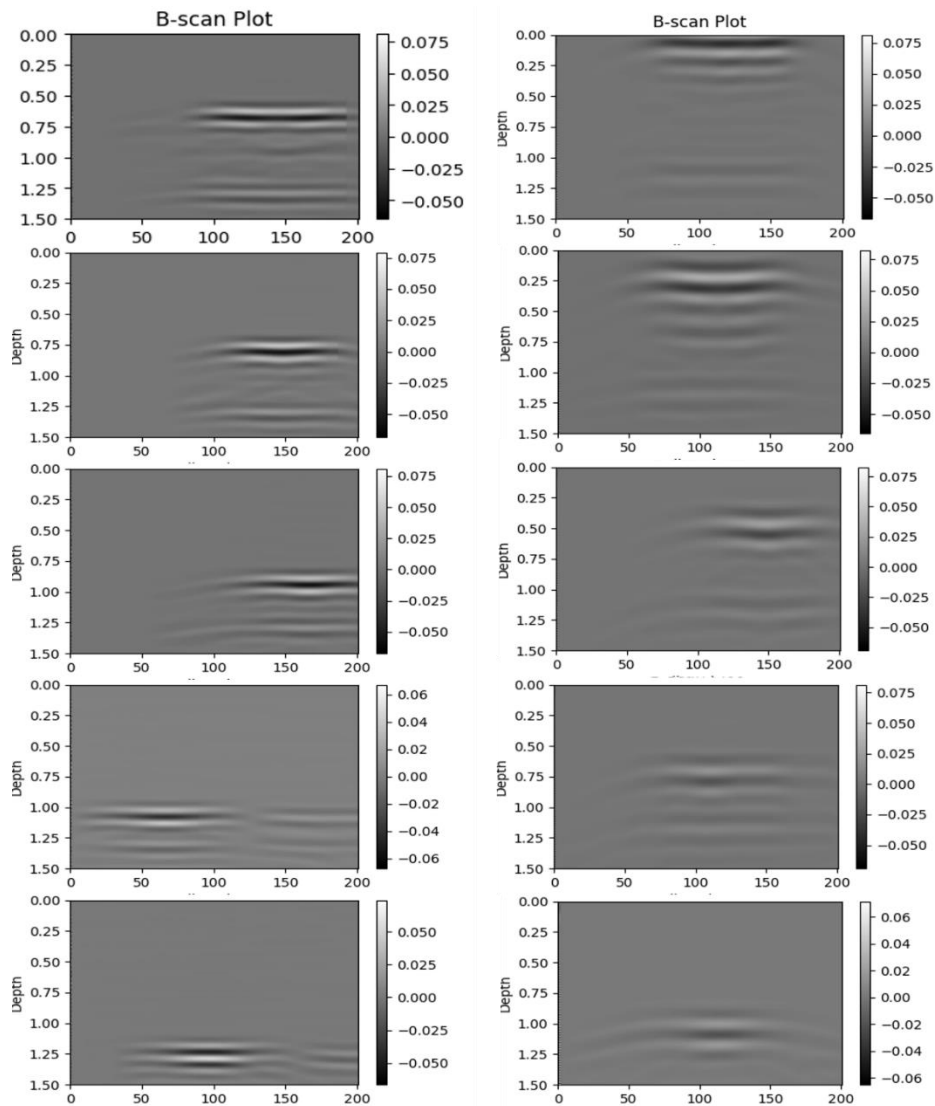


Fig 48. Identification of the differing layers present in model crevasse from both antennae, Vivaldi (left) and V14 (right)

The results in Fig 48 did however indicate that the Vivaldi antennas showed a more pronounced reflection, making the detected foil clearer in the B-scans. The next tests involved modelling an “inverse crevasse” by placing dielectrically similar material to ice—within a void surrounded by Styrofoam (see Fig 32). When taking a B-scan of this model, the resulting image showed a paraboloid-like curve, as expected from a crevasse. Fig 49 shows the resulting B-scans when placing the clay within the model. These results were a major success for the group as they provided experimental evidence that crevasses could be detected. Furthermore, the parabola within the B-scans were detected as crevasses when ran through the ML algorithm, more details on these developments will be expanded on later in the report. Though many advancements had been made by the group, these results provided the group with new problems that needed to be addressed. For instance, the fainter line seen directly below the parabolic curve is in fact the table and was an artefact of the current background subtraction algorithm, indicating that more work should be done on improving the algorithm.

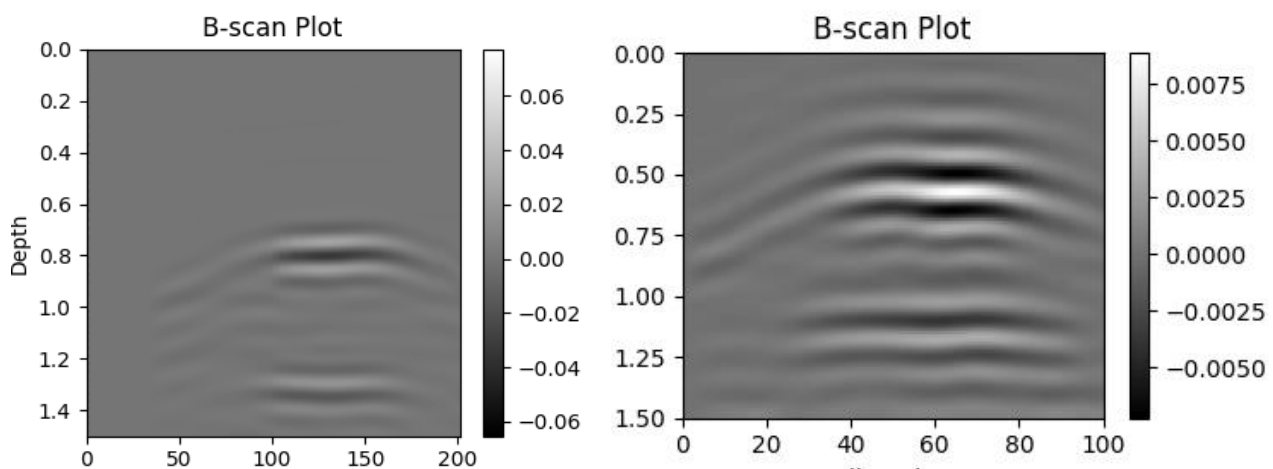


Fig 49. “Inverse crevasse” model showing parabolic curve detected by Vivaldi (left) and V14 (right) Antennas.

2.7. Testing the Automated Detection

The Machine learning crevasse detection shows successful detections on B-scans generated by the SSPOC with the synthetic testing medium, with a high rate of success and detection confidences over 0.50, as seen in the example in Fig 50. However, a comprehensive test to ensure a success rate of 75% as specified in the initial objectives is still pending. It would be carried out by performing a large number of tests (40 tests) for different orientations, volumes and positions of the clay insert, and aiming for 75% of tests (30 tests) or more yielding automated crevasse detections with confidences of more than 0.25. This would provide a strong guide for users to assess the safety of a given area and judge the presence of a crevasse.

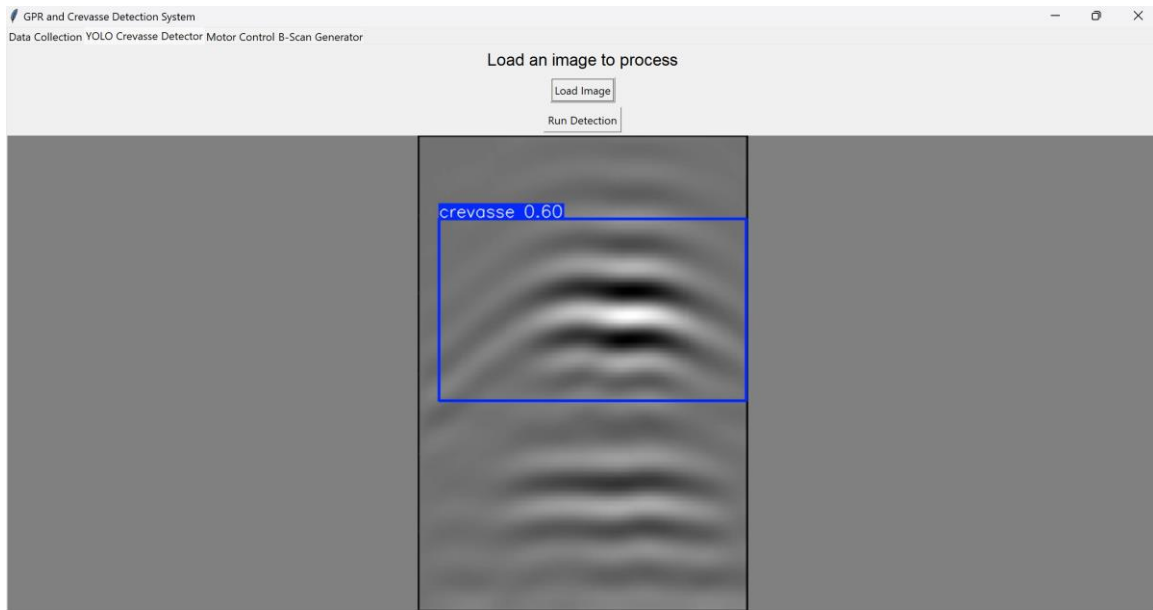


Fig 50. *Successful automated detection of model crevasse by machine learning*

2.8. Theoretical Validation

Due to budget constraints and a lack of an adequate testing area to better simulate real conditions, further practical tests on the SSPOC were no longer possible. However, the group carried out theoretical tests to conclude the legitimacy of the groups GPR system when operating over an expected real crevasse. The group set the goal of having a system capable of detecting crevasses hidden beneath a 6 m snow bridge. The group used the radar range (13) to deduce what power would be required to be transmitted such that a crevasse of the defined depth could be detected. Further assumptions had been made on the characteristics of snow and ice based on findings from academic journals. Due to the dry climate of polar regions, it was assumed that the type of snow present within these regions will be exclusively dry snow. As such from a study done in [62], it is reasonable to assume that dry snow is fully transparent for snowpacks that are smaller than 300 m in L-band emission.

while another study found that the attenuation of polar snow at a frequency of 380 MHz to have also virtually no attenuation [63].

Therefore, for the calculation of the depth penetration of signals the minimum assumed attenuation would be 0.001 dB/m, while a maximum attenuation of 1 dB/m was also included in the calculations as contingency if the prior assumption was incorrect.

$$P_e = \frac{P_s G^2 \sigma \lambda^2}{(4\pi)^3 R^4} \quad (13)$$

For the GPR prototype to be able to detect a crevasse of a depth of 6 m, when hovering over the surface at a height of 1 m, the power output of the system would need to be considerably higher than that of the output of the Anritsu. Assuming the low end attenuation of snow, the V14 antennas would need an output power of 0.7 W to detect crevasses of such a depth. Contrastingly, the Vivaldi antennas would require a far higher 25 W of output power for detection. These outputs would rise to 10 W and 300 W respectively, when assuming the maximum attenuation of the snow bridge. These findings suggest that the V14 antennas would be better suited to the final product.

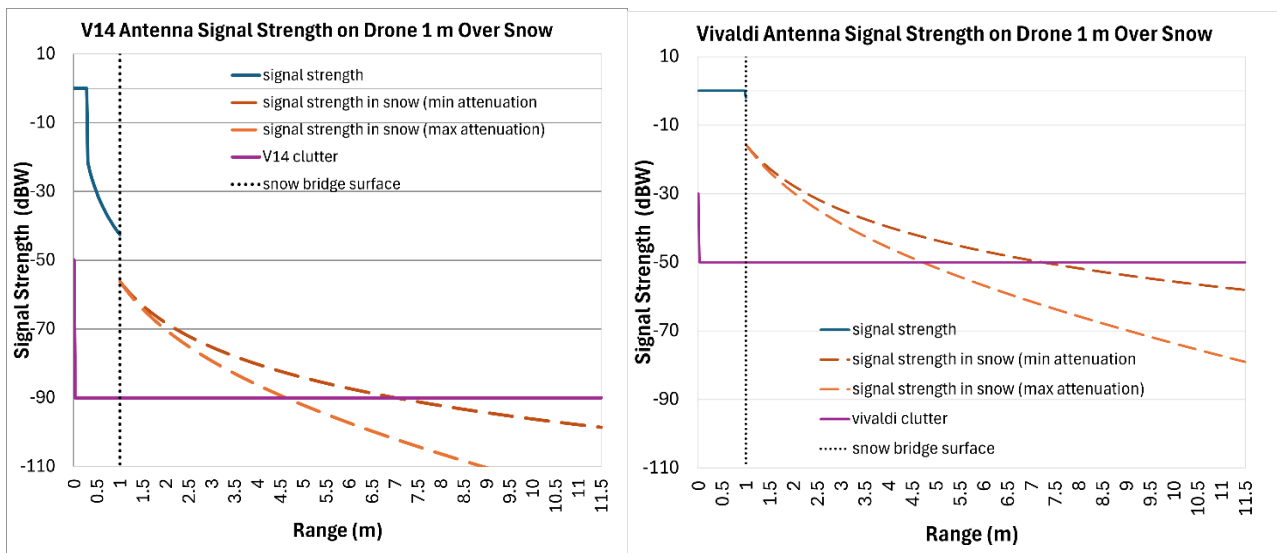


Fig 51. Theoretical signal strength intensity change as the waves travel through the snow bridge.

In the event of an ice bridge forming over a crevasse, it is worth noting that the depth needed to travel to detect a potentially harmful crevasse would be far less than through snow. In fact, an ice bridge thickness of just 0.38 m would be able to withhold a weight of 5000 kg [64]. The results of the depth penetration of the signals in ice of the V14 and Vivaldi antennas are both 1.6 m when operating at 0.7 W and 25 W, respectively. These results would indicate that this system should detect any unsafe ice bridge, further proving the potential usefulness of this technology.

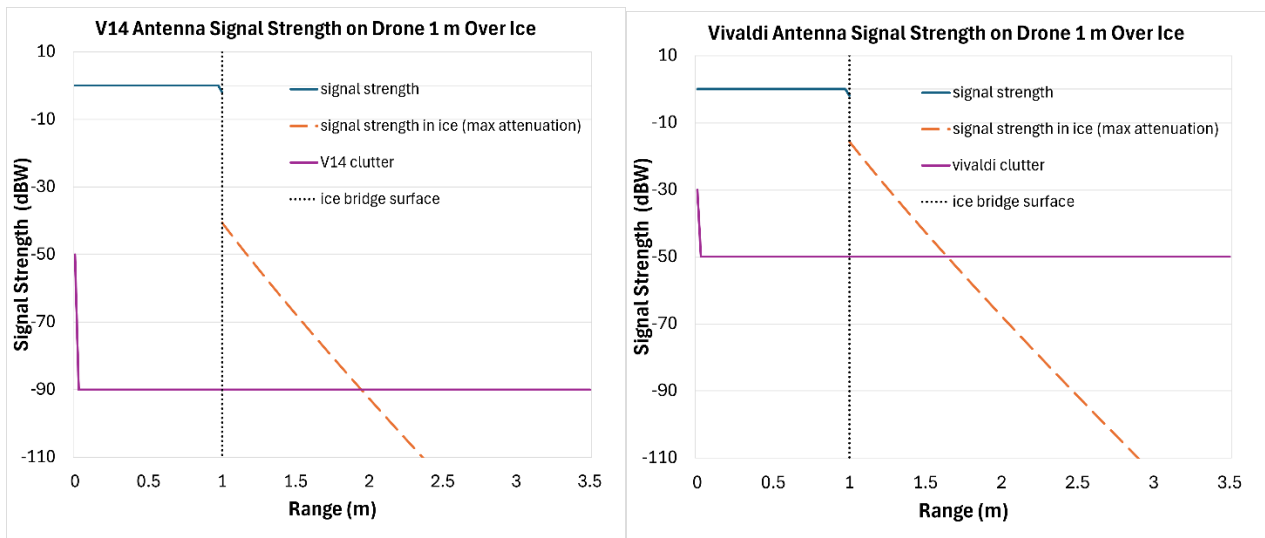


Fig 52. Theoretical signal strength intensity change as the waves travel through the ice bridge.

These calculations indicate that the V14 antennas would be preferred for usage in the final system. It is worth noting however that both antennas are not optimised for this GPR system, but they do suggest that the loaded dipole set up of the V14s may prove to be a better option for the project. Future iterations of the project could aim to create system specific antennas optimised for the project – research that goes beyond the scope of the project in its current form.

3. Project Management

3.1. Technical Objectives

In Semester 1, the project's technical objectives were primarily about understanding the scope, doing background research and preparing the foundational deliverables of grant pack and project pitch. Considering the initial guidance, approximately 25% of the overall workload was allocated to

Semester 1 and 75% was planned for Semester 2. Consequently, Semester 2 had a significant increase in terms of both the complexity and volume of technical tasks including detailed subsystem designs, extensive testing, and system integration. This caused a large increase in working hours and it required more of coordination and commitment from the team members.

3.2. Meetings and Communication

Keeping the consistent communication was pivotal for managing the complexity of the project. Throughout both semesters, the team maintained a structured routine including at least one team meeting and one supervisor meeting in each week, with few exceptions. Monday meetings were specifically organized to see and criticize progress from the previous week and to outline individual tasks for the next week. This helped in terms of accountability.

Each supervisor meeting was provided by an agenda document prepared and sent by the Secretary. Minutes from these meetings were also documented and shared after meetings. This ensured clarity among team members and supervisors.

Several communication platforms were used to control team communication:

- **WhatsApp** [65]: Used as the main channel for quick updates, reminders and day-to-day team communications.
- **OneDrive** [66]: Used as the repository for project-related documentation. It was structured to include folders for literature reviews, technical documentation, meeting agendas, minutes, and deliverables. This provided easy access for all team members and supervisors.
- **Teams** [67]: Used for all remote meetings, including stakeholder meetings, meetings for outreach and several team meetings like those for updating documentation or to accommodate teammates not on site.
- **Connecteam**[68]: Utilised the geo-locked clock in and clock out feature to boost accountability and support equitable workload distribution.

3.3. Planning and Task Allocation

Initially in the project planning, a provisional Gantt chart was created early in the first semester. However, this initial chart was not detailed, and it was not extensively utilized in the first phase. It was recognized that there is a need for a better organization, a more comprehensive Gantt chart was made during the second semester. This updated chart was more effective. It clearly outlined major milestones, technical deadlines, and deliverables.

In Semester 2, other visualization tools were also used, including a shared PowerPoint slide and a physical whiteboard. Weekly responsibilities, completed tasks, and pending assignments displayed there. This helped in terms of efficient task management, minimizing miscommunication and improving accountability.

Group roles were assigned on the basis of preference and interest, for example: signal processing lead, software lead etc. These were followed through and ended up being exactly how the work was distributed, hence confirming the effectiveness of the approach. Those teammates without a formal role (such as auditor, manager or secretary) ended up taking up slightly more of the technical work to equitably distribute workload.

3.4. Budget Management

A detailed budget management was executed under the auditor's supervision. The auditor also verbally updated team members and supervisors about financial status and budget allocations. Resources were efficiently utilized in line with the needs of the project and the deadlines. The budget was managed on an Excel spreadsheet and documented every spend, order dates and the dates when items were received. This document can be seen in Appendix C.

The original budget of the project was £1,650.00 and this was initially divided up to have £882.00 for building the bespoke GPR prototype, £612.00 on the UAV and £149.40 as a 10% contingency, leaving £6.60 spare, as can be seen in Fig 53 (left). It was quickly decided to remove the UAV from the project scope, allowing for more funds to be spent. This was useful as it allowed for higher specification electronics to be bought to build the GPR with and for items to be bought to build the GPR test rig (which was not originally allocated in the budget). The final spending was £1,503.47, leaving £142.58, close to the original total predicted spend. The spending categories differed, with £1,152.07 being spent on the GPR, £319.28 on the GPR test rig and £36.07 on antennas, as shown in Fig 53 (right).

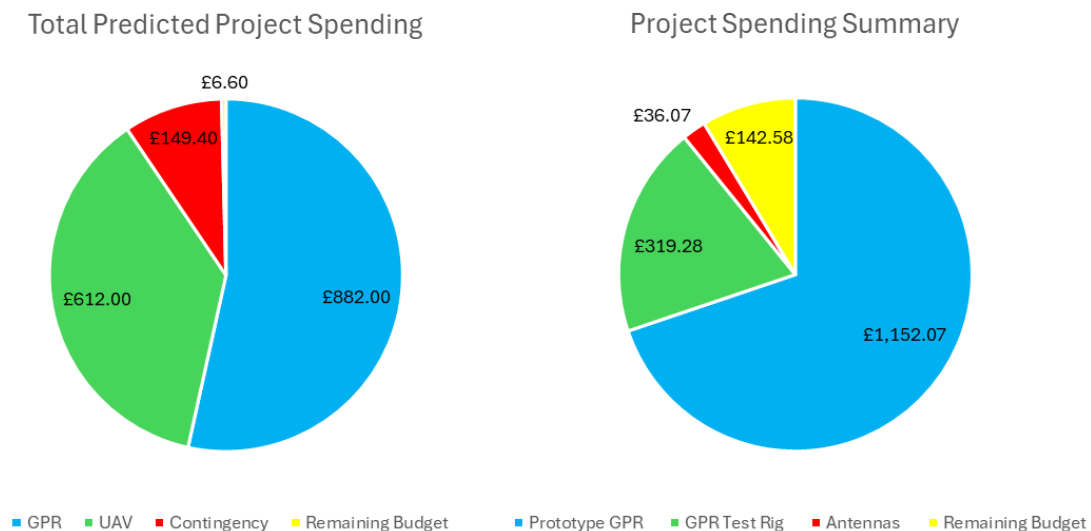


Fig 53. Project budget. Predicted spending (left) from the beginning of the project and total spending (right) from the end of the project.

3.5. Team Behaviour, Dynamics and EDIA

The team has been working in a respectful and collaborative working environment. Being open when communicating allowed team members to offer mutual support. Each member provided assistance to each other whenever needed. While task distribution was aimed to be equal, some flexibility was applied for accommodating individual circumstances. These were remote working necessity, international travels, or absences related to health.

The project team upheld EDIA principles by recognizing and respecting expertise diversity, different experiences and cultural backgrounds of members. Language barriers, influencing the members having English as a second language, were actively managed by providing clarifications or repeating the word when necessary. Religious observances were respected and accommodated. These practices provided an inclusive, respectful and comfortable working environment for all members.

3.6. Outreach and External Engagement

Engaging with external experts and industry professionals contributed to project's success significantly:

- **David Daniels CBE:** Provided invaluable technical guidance and insights. He influenced the project's technical direction, especially about antenna selection and testing strategies.
- **Daniel Vale and Workshop Staff:** Provided essential assistance in hardware tasks, practical assembly and prototyping.
- **Raun Khangerey:** Shared his expertise from his research project which is parallel with this. He helped shaping decisions about radar technologies and antenna optimization.

- **Eef van Döngen:** Offered practical insights related to Arctic research conditions. This contributed to group's contextual knowledge for deciding system design considerations.
- **George Brown:** Gave feedback on the theory of the circuit designs for the bespoke GPR prototype, reducing the risk of the circuits not working properly. This was important as the first parts order cost over half the original budget.
- **EEEOps Team:** Helped with placing orders and getting deliveries.

These collaborations underscored practical relevance and technical robustness of the project.

4. Overall Summary of the Project

4.1. Achievements Against Aims and Objectives

The project was guided by an overall aim and five objectives that fit within the scope and technical constraints of the whole system. Each objective was carefully tracked and addressed within development and testing. A summary of achievements against these aims is provided below.

The first objective was to develop a comprehensive small-scale proof of concept using a Vector Network Analyser (VNA) on a repeatable and robust testing rig. This objective was successfully met. A modular platform was constructed, which allowed consistent and repeatable testing and results, and it yielded meaningful radargrams. The output demonstrated subsurface reflections and crevasse features, validating the feasibility of the concept.

The second objective involved developing a robust signal processing pipeline for radar data acquisition and interpretation. This was done with a full integration from hardware signal capture to user interface display. Both A-scans and B-scans were implemented, with B-scans offering real-time improvement options like background subtraction and inverse-square scaling. These features allowed the operator to improve the visibility of key reflections.

The third objective was to implement an automated machine learning analysis pipeline with a target of over 75% accuracy. Even though the full validation is still pending, early implementations are promising. The model was trained on a set of radargrams with known ground truths, and preliminary outputs demonstrate credible detection performance. Quantitative metrics such as precision and recall will be assessed in future project extensions.

The fourth objective involved validating system performance with simulations and practical testing, which was also addressed. Theoretical calculations were used to set the antenna separation and power requirements, while hands-on validation with both the bespoke GPR unit and the VNA-backed testbed confirmed signal fidelity, usability, and reliability. Radargram clarity and component integrity remained stable throughout multiple test runs.

Finally, an objective was to develop a compact and bespoke GPR prototype that is suitable for UAV-mounting. This goal was also achieved. A radar system was built with a measured weight below the 5 kg constraint and operated using a 12V DC power supply, suitable with UAV integration requirements. The prototype has a flexible foundation for further testing in field-like conditions and eventual airborne deployment, and only requires an enclosure. It successfully produces A-scans, however, B-scan processing is still needed.

To visualise the objective completion status, the table below summarises each aim alongside a justification.

Table 9. *Summary of Project Objectives*

Objective	Achievement
Do a comprehensive small-scale proof of concept with complete test results with an existing VNA, complete with a repeatable, robust testing rig.	A small-scale proof of concept was executed, on a robust rig which can repeatedly be demonstrated. The produced radargrams were meaningful and indicative of successful detection.
Develop a robust signal processing pipeline for radar data.	Data acquisition from the hardware all the way to the UI was performed and displayed as both A-scans and B-scans. B-scans had the option for enhancement techniques such as background subtraction, inverse squaring magnitude, etc.
Implement a machine learning algorithm for automated analysis and achieve over 75% accuracy.	The automated detection is yet to be formally assessed. Hence an exact precision figure is pending, however shows good promise in detection on tests so far.
Validate system performance using simulations and testing.	Comprehensive testing and theoretical confirmations validate system performance.
Develop a compact, bespoke GPR prototype suitable for deployment on a UAV.	The GPR prototype demonstrates successful basic functioning with a weight much under 5 kg and within a required power of 12V DC.

4.2. Progress Against Plan

The project followed a dynamic timeline that evolved significantly between the initial plan and final implementation. Semester one was intentionally designed to carry around 25% of the workload, focusing on background research, literature review, grant pack submission, and stakeholder outreach. A provisional Gantt chart was drafted during this stage, but it was not maintained or referred to consistently, reflecting the exploratory nature of the early work. Technical work in semester one was limited to GPR simulations and drone control simulations as well as front-end UI development.

Semester two carried the bulk of the technical progress, and a new Gantt chart was created following a critical turning point in the project: the meeting with Professor David Daniels. His expert feedback highlighted the limitations of pursuing an Arctic simulation with constrained resources and advised pivoting towards a realistic laboratory-scale testbed. This insight led to a strategic decision to shift the emphasis from real-world deployment to internal subsystem validation. Shortly after this, a new planning session was held to reallocate tasks and create a clearer and more actionable timeline. From then on, weekly targets were tracked using a whiteboard and a shared PowerPoint slide documenting task ownership, previous work, and upcoming goals. This approach ensured visibility and accountability within the team and allowed adjustments in real time.

While the workload intensified, particularly during blocks 4 and 5, the team adapted quickly. For example, data acquisition tests using the Vivaldi antennas and signal processing chain were completed on time, even as other goals such as drone testing and final ML integration were shifted beyond the initial timeline. The structured planning process and adaptability of the team played a crucial role in navigating changes without compromising the core objectives.

4.3. Suggested Future Development

The current testbed provides a strong foundation for several avenues of future work. The most immediate priority is the expansion of the radargram dataset. More extensive simulations, involving diverse subsurface geometries, snow densities, and noise profiles, would enable training of a machine learning model capable of identifying crevasse boundaries in real time. These datasets could also be used to benchmark the system against traditional GPR outputs and validate accuracy improvements. With further refinement, an onboard inference module could be developed using a lightweight microcontroller or edge processor to process data mid-flight, minimising latency and improving field usability.

Hardware refinements should also be considered. Antenna separation proved to be a limiting factor, particularly with the larger Vivaldi designs, which require an aperture size that is difficult to implement on compact UAVs. Further research into miniaturised wideband antennas and the application of electromagnetic shielding or balun structures could reduce mutual coupling and increase detection accuracy.

Finally, UAV deployment must be explored. The effects of dynamic motion factors such as drone vibration, altitude variation, and angular tilt on signal quality are still unknown. A future field test in a controlled environment, such as a glacier simulator or open terrain with buried targets, would enable full validation of the system under flight conditions. Finally, although system validation through simulation has been briefly explored, thorough and documented validation is still required.

4.4. Final Conclusions

The project has delivered a modular, lightweight, and power-compliant GPR system designed for UAV deployment in crevasse detection applications. Despite initial ambitions of real-field deployment, a mid-project pivot allowed the team to concentrate on lab-based subsystem testing, which significantly improved feasibility and realism. This pivot, informed by expert feedback, was a major contributor to the project's success. The project also delivered a comprehensive small-scale proof of concept complete with a signal processing pipeline and automated crevasse detection.

Core aims related to hardware design, antenna testing, GUI development, and system integration were achieved. Although thorough machine learning testing and real UAV integration remain open tasks, their groundwork has been established. The project is now positioned for rapid extension into those areas with minimal redesign.

Ultimately, the team has produced a reliable and scalable platform that addresses a real-world safety problem with promising potential. With continued refinement, the system could be deployed in the field to enhance safety in polar exploration and research, reducing the risks posed by hidden crevasses while offering a cost-effective and flexible alternative to current detection methods.

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